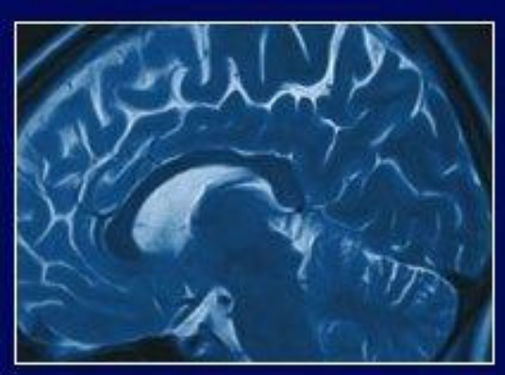

The Waisman Laboratory
for Brain Imaging and Behavior

University of Wisconsin
SCHOOL OF MEDICINE
AND PUBLIC HEALTH

Topology for Image Analysis

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Spectral geometry

Trees

Segmentation

Tubular structures

Clustering

TDA

Persistent homology

Topological constraints

Topology correction

Why we need topology in image segmentation?

Image segmentation is a topological operation of making an image into multiple disjoint regions. The goal of segmentation is to simplify and/or change the representation of an image (functional data) into discrete states.

A stroke patient with dysphagia in a diffusion tensor image

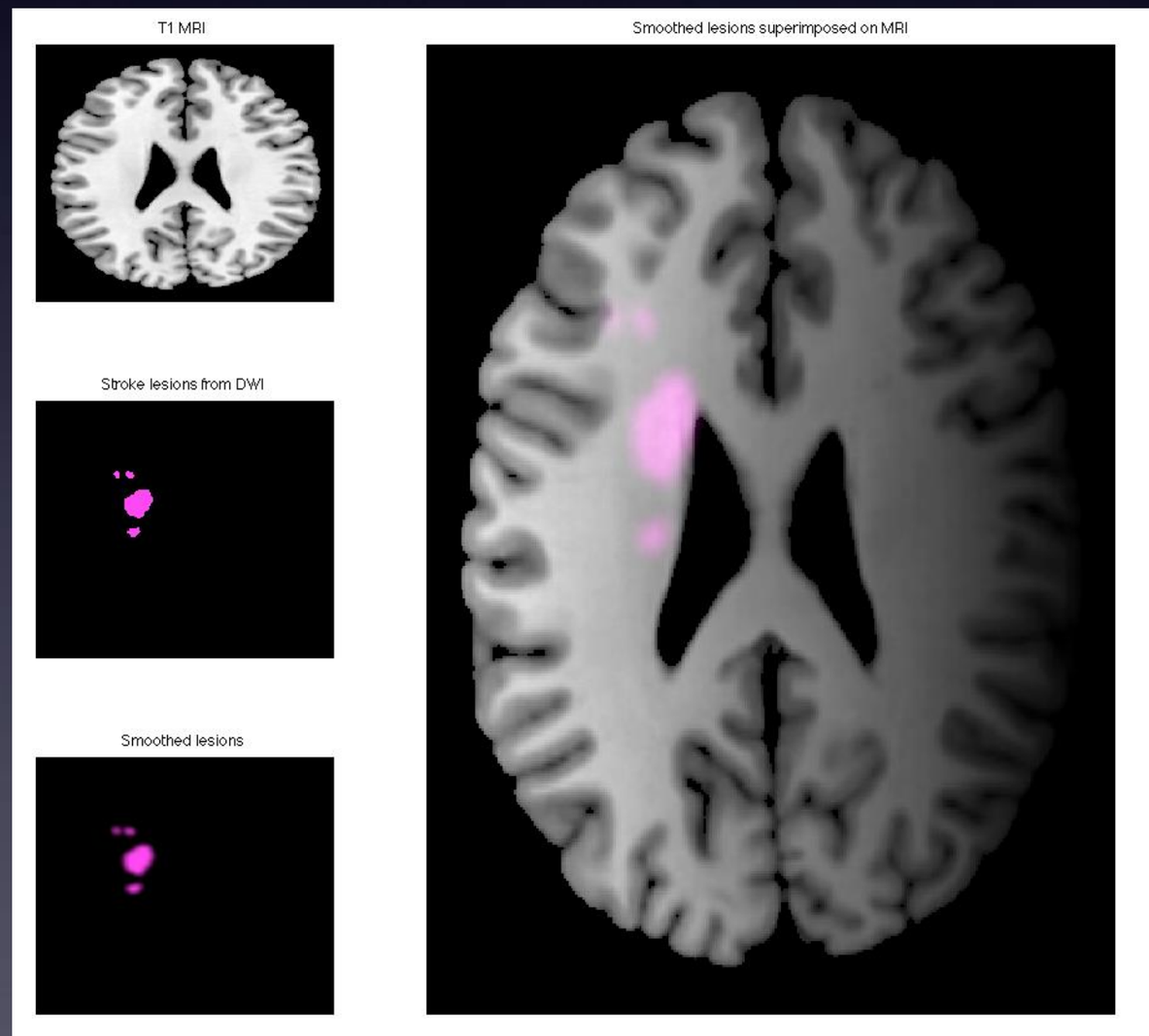
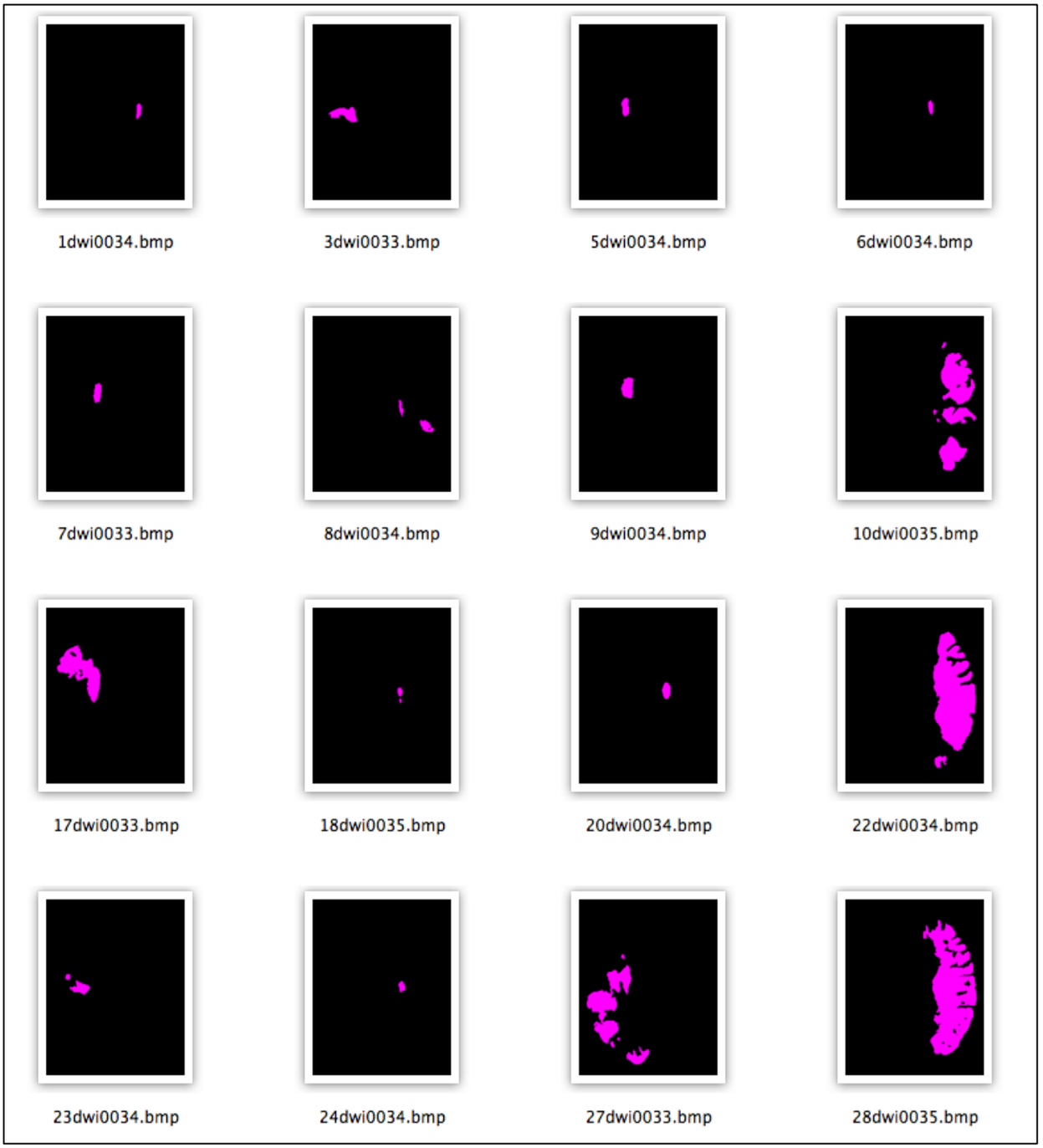
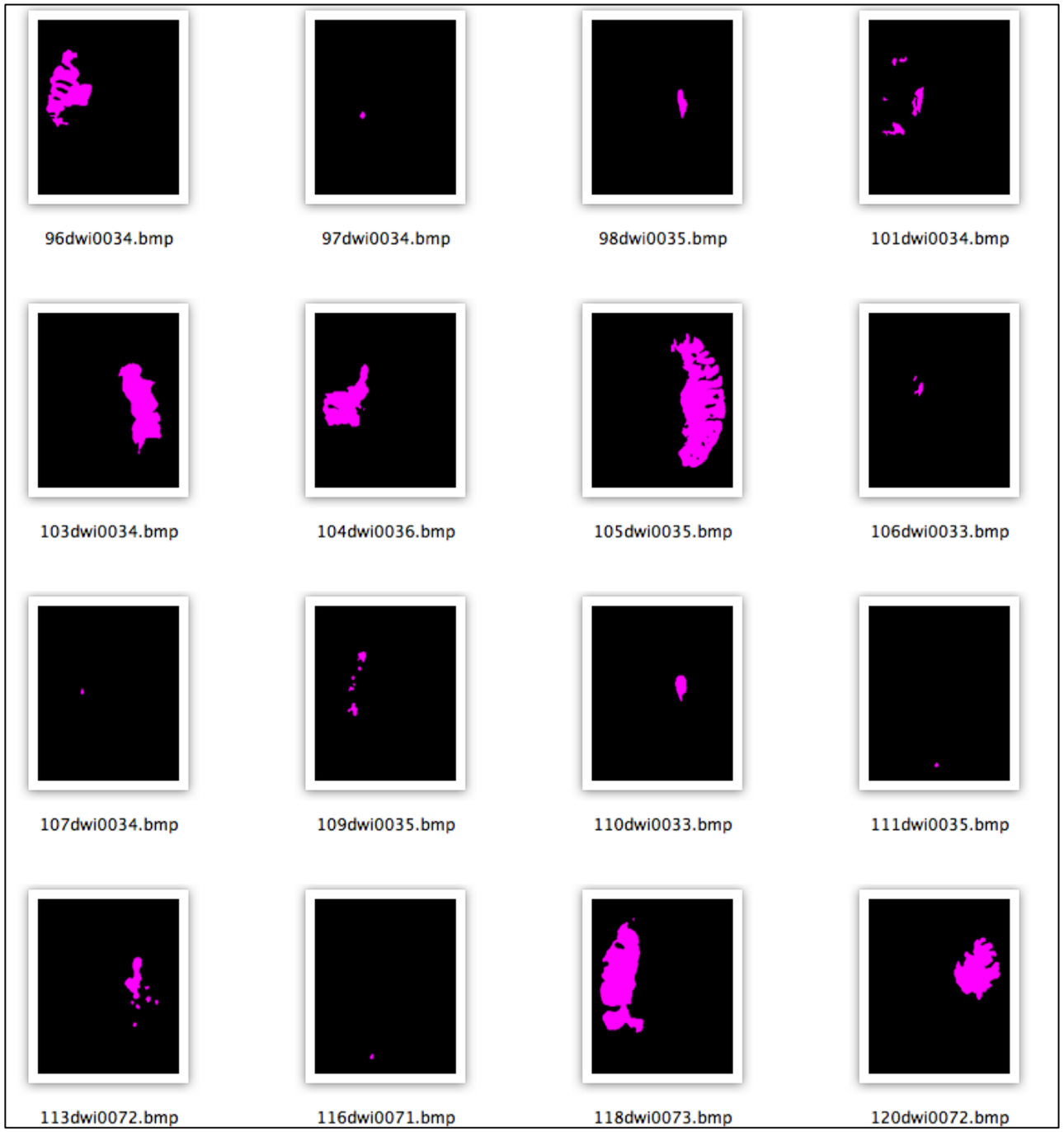


Image credit: Dong-Eog Kim, Dongguk University, Korea

Stroke patients with dysphagia



Group 0: patients who got improved after one month (n1=58)



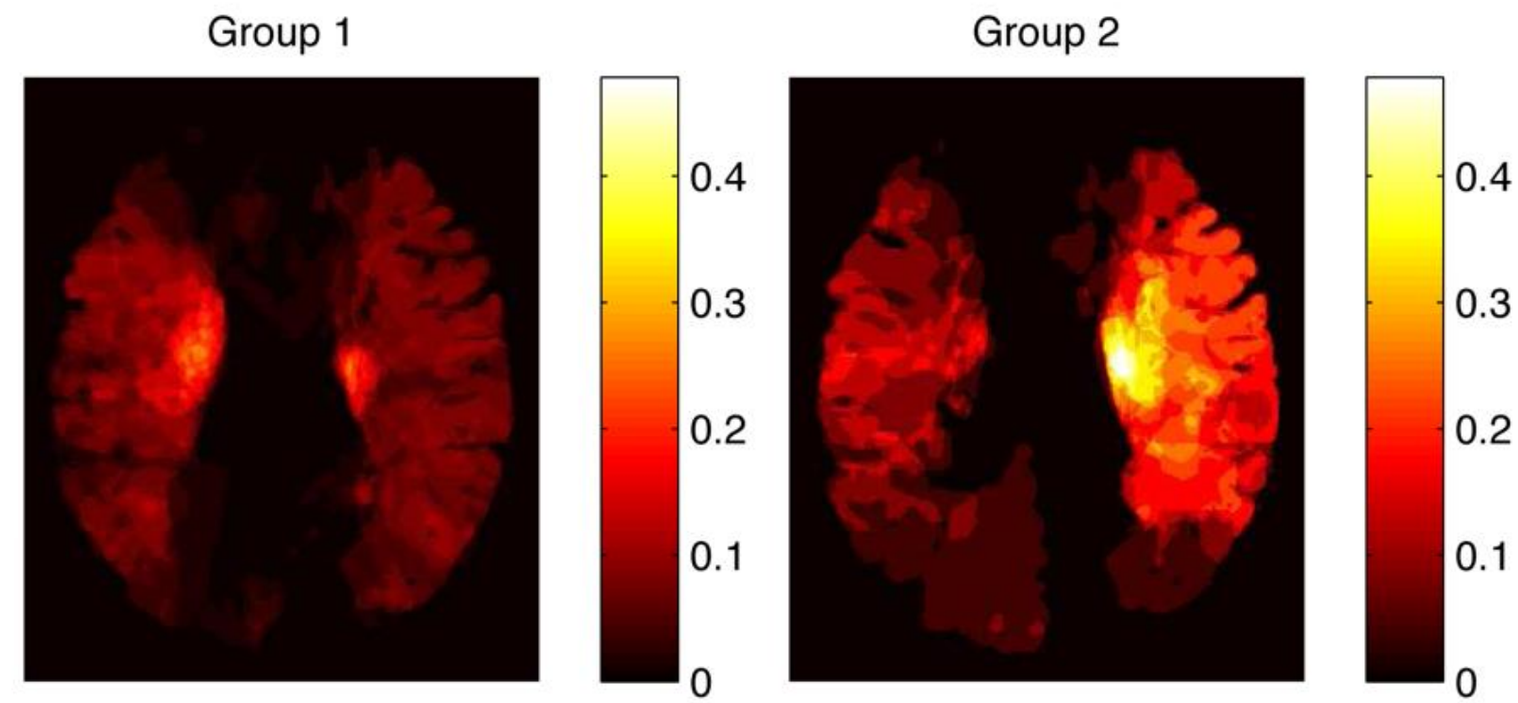
Group 1: patients who didn't get better (n2=23)

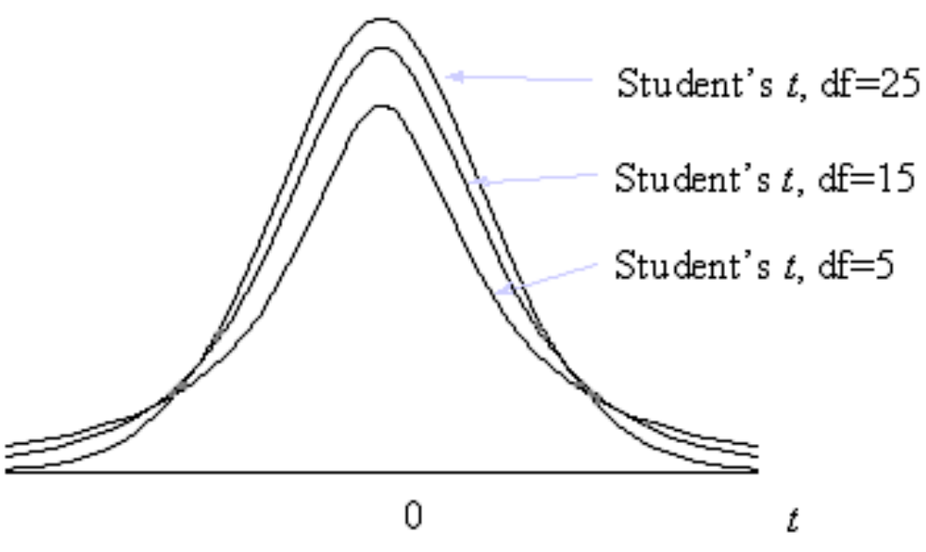
Question: How we discriminate the groups?

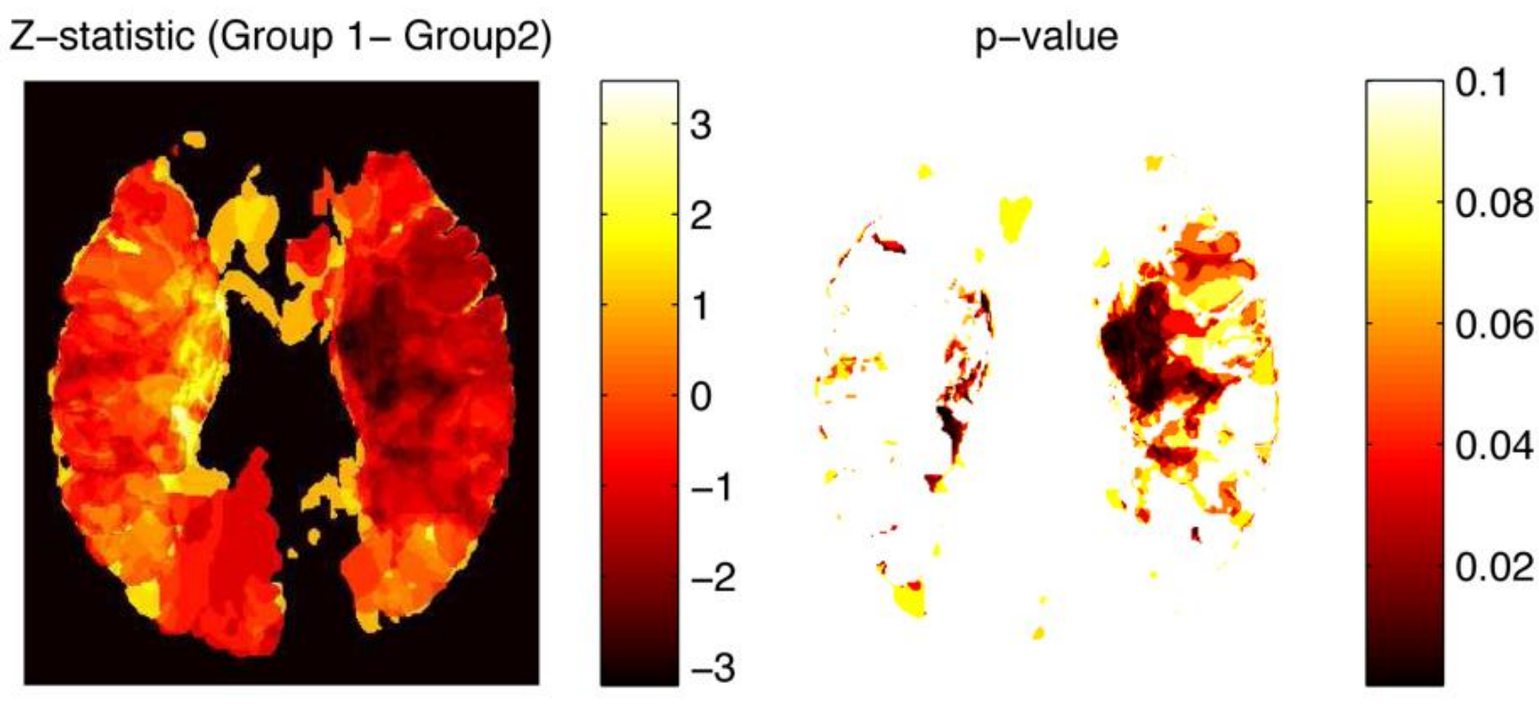
Two sample test at each pixel without smoothing

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim t_{n1+n2-2}$$

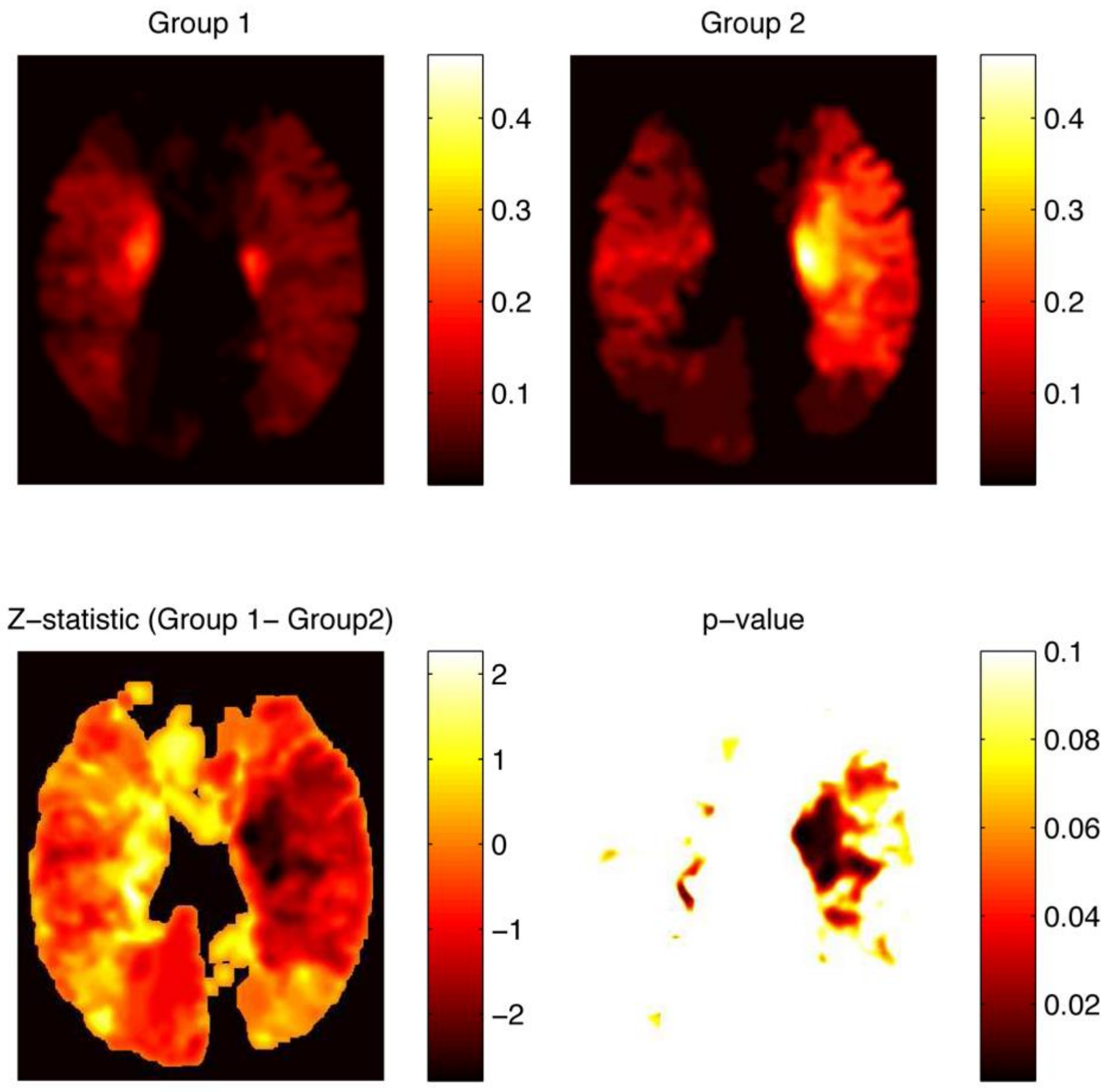
$$s^2 = \frac{\sum_{i=1}^{n_1}(x_i - \bar{x}_1)^2 + \sum_{j=1}^{n_2}(x_j - \bar{x}_2)^2}{n_1 + n_2 - 2}$$



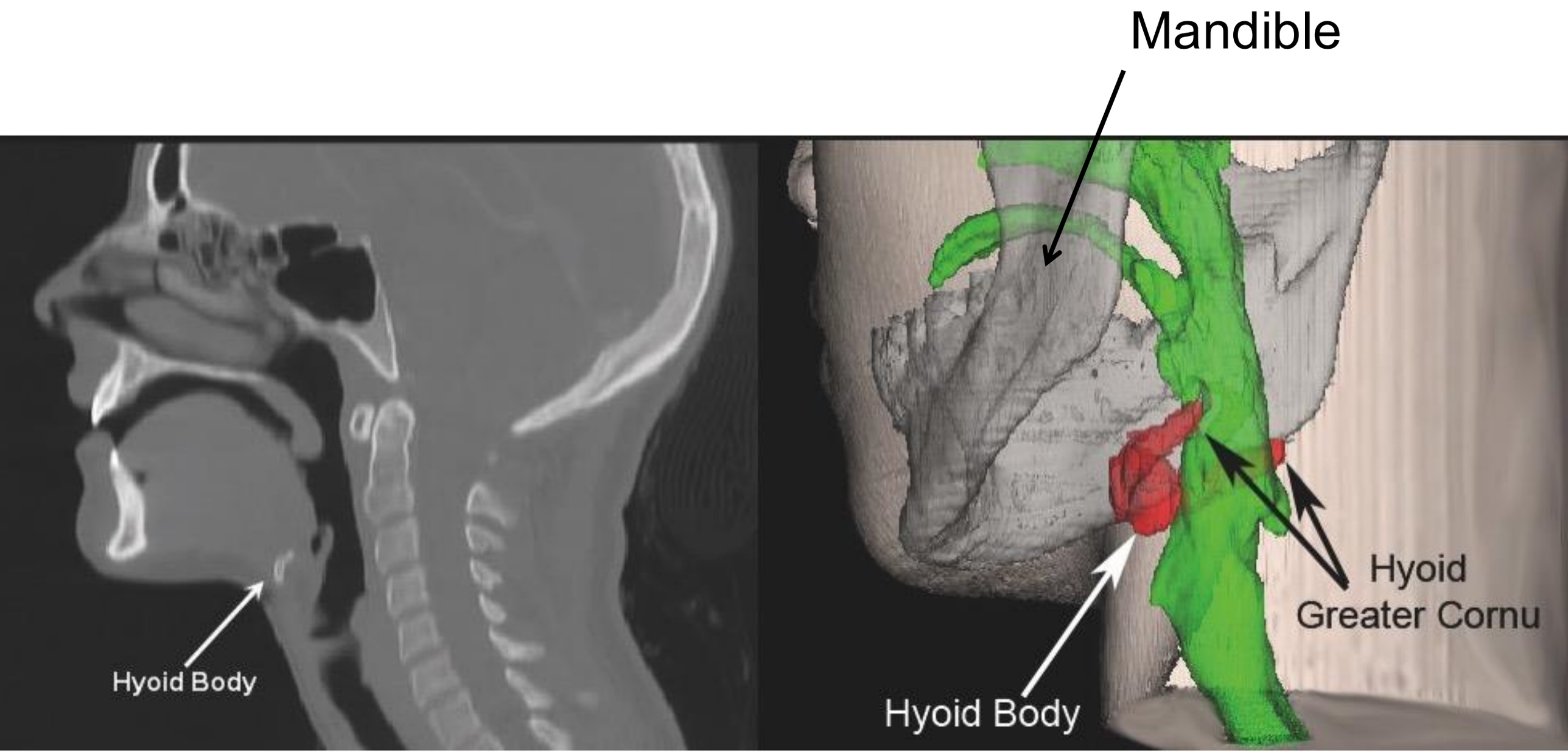




Two sample test with smoothing



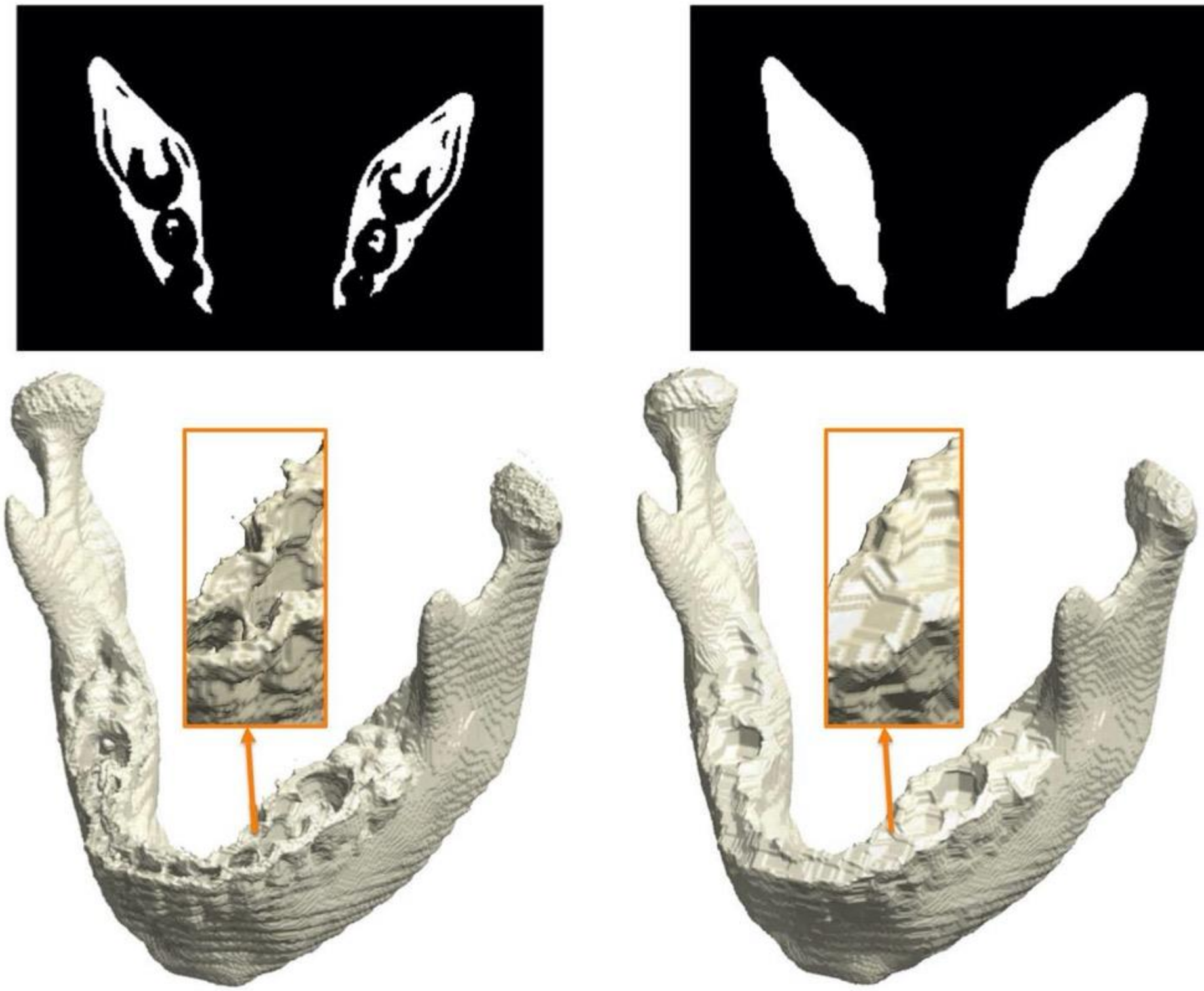
Head and neck CT image



Semiautomatic histogram thresholding in ANALYZE package

→ Automatic pipeline using ANTS-based template matching

Topology correction in segmentation

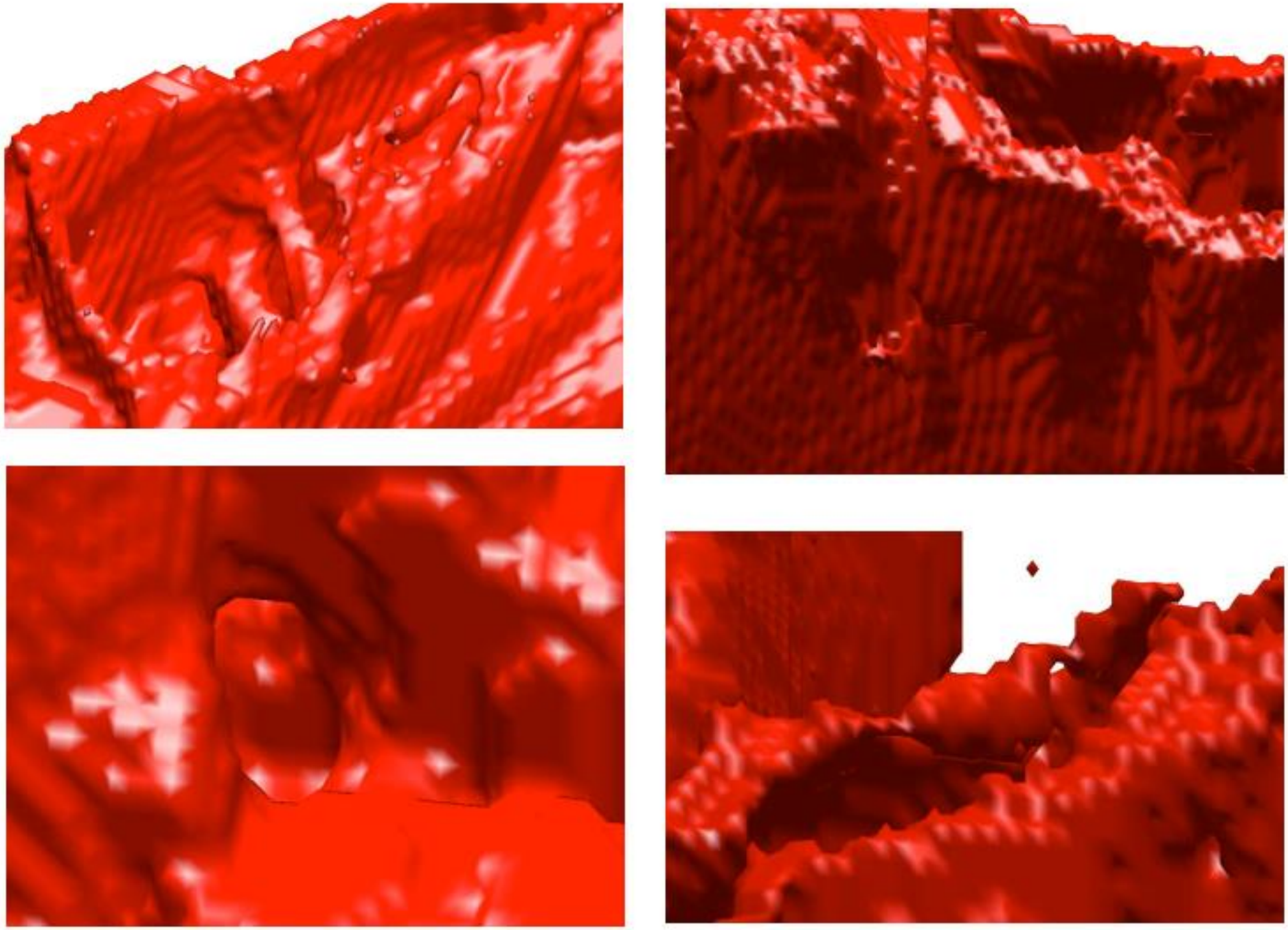


Hole & handles corrected using Euler characteristic

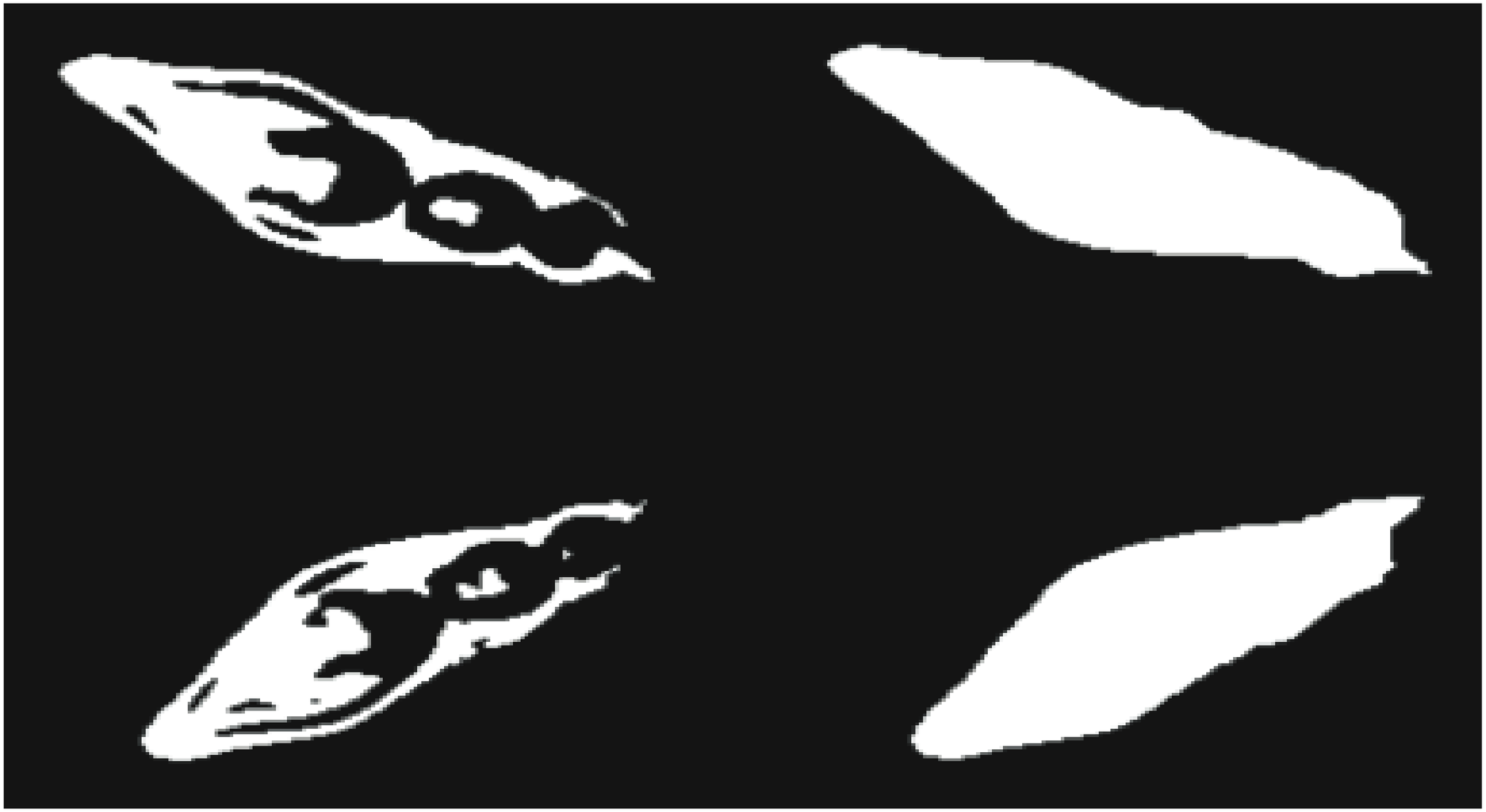
Chung et al. 2015 Medical Image Analysis. 22:63-76

Euler characteristics

How do check if topological defects in images are corrected without seeing images?

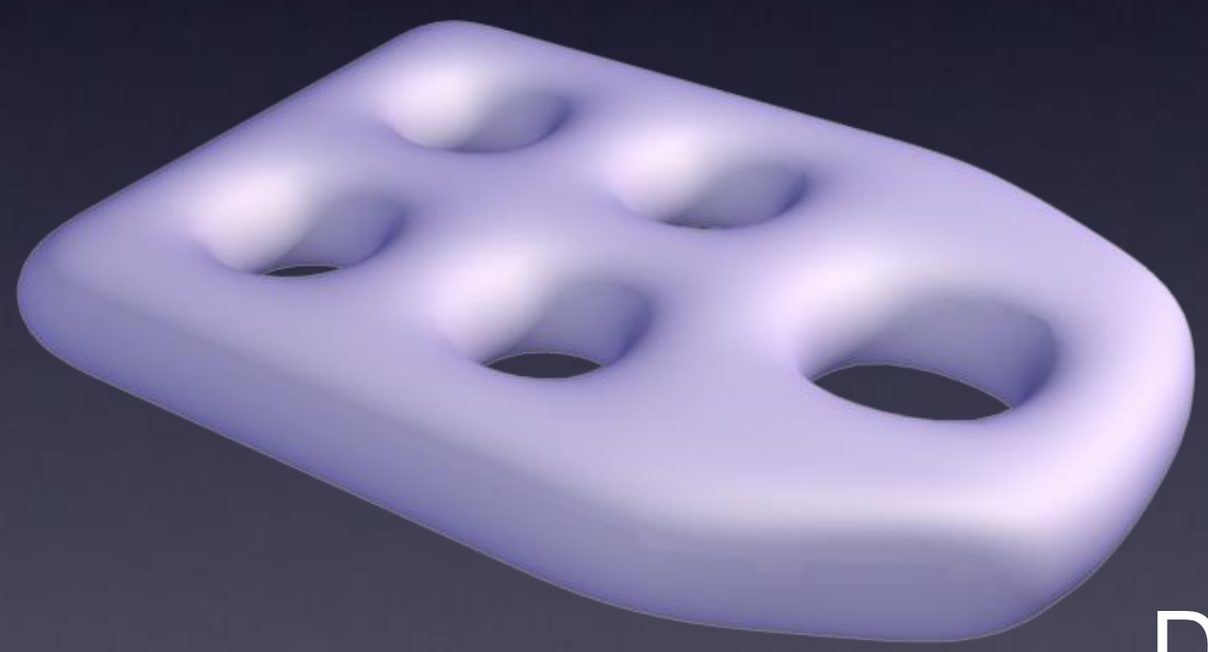


Volume rendering of a mandible from CT
Holes and handles in binary volume

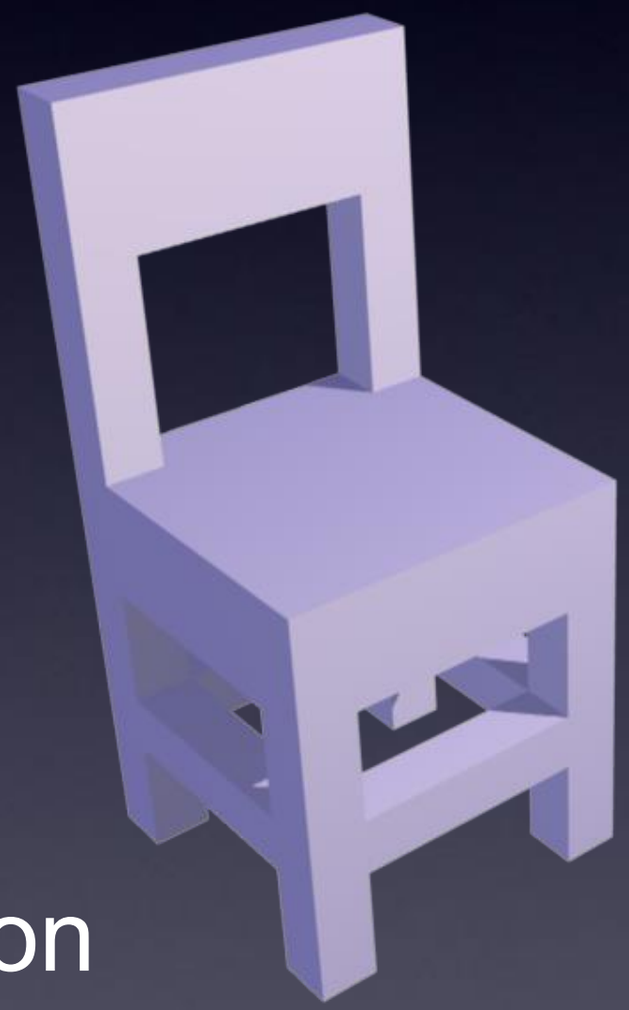


By checking the Euler characteristic of the binary volume of a mandible, holes in the binary volume can be detected. This process is necessary to make the mandible binary volume to be topologically equivalent to a solid sphere.

Genus = the number of holes in a surface or an object.



5 holes



Deformation invariant

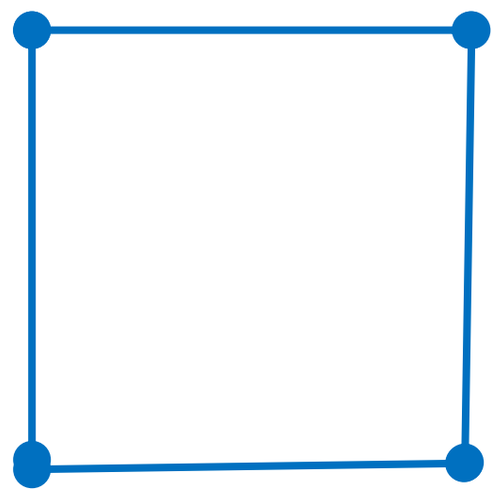
Image credit: Matt Black

Euler characteristic with convex polyhedrons

Polyhedron is a solid object with polygonal faces, edges and nodes.

Euler characteristic: EC = N – E + F - … .

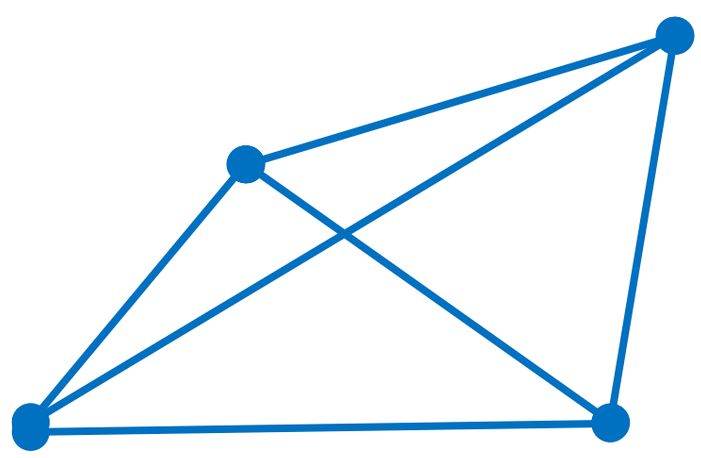
$$EC = N - E + F$$
$$= 4 - 4 + 1$$
$$= 1$$



If there is no face

$$EC = N - E$$
$$= 4 - 4$$
$$= 0$$

$$EC = N - E + F - V$$
$$= 4 - 6 + 4 - 1$$
$$= 1$$



If there is no volume

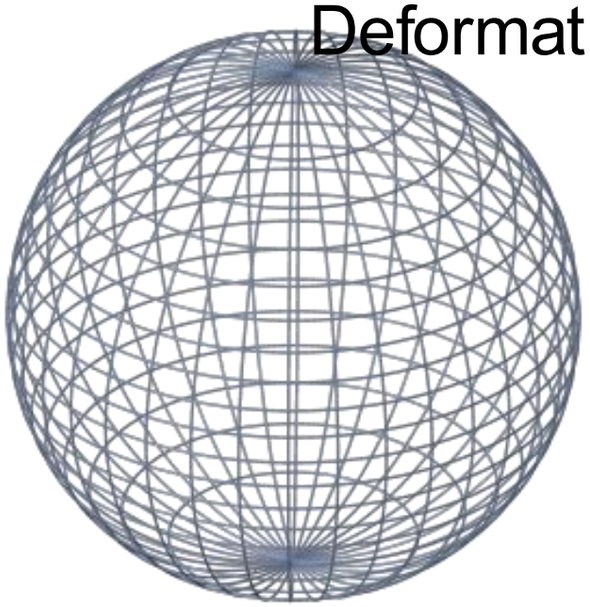
$$EC = N - E + F$$
$$= 4 - 6 + 4$$
$$= 2$$

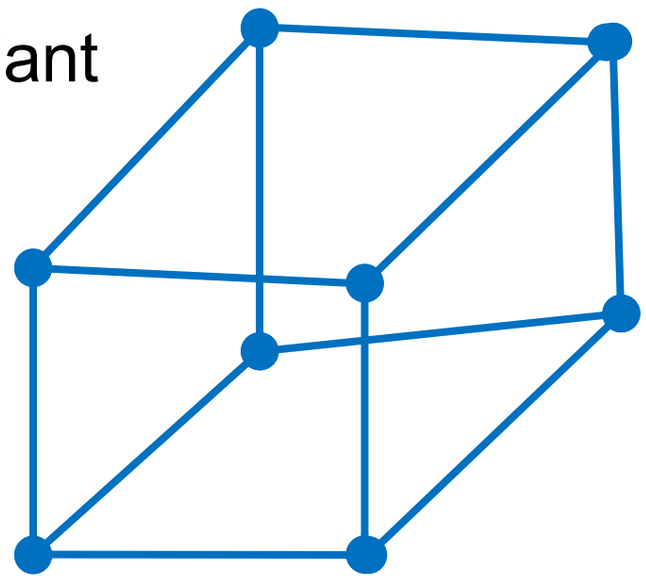
Question: Check if we have EC = #dimension - #hole

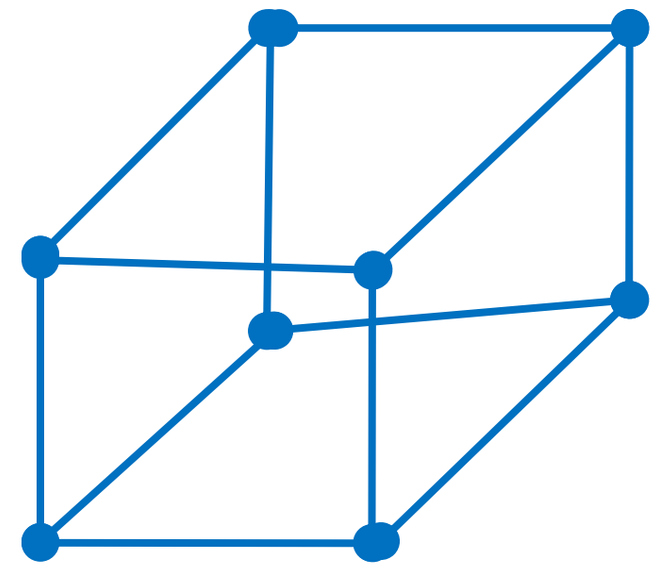
EC is an approximate measure of data dimension!

Computing Euler characteristic of 3D objects

Topologically
equivalent
Deformation-invariant







Sphere

$EC = N - E + F$

$= 8 - 12 + 6$

$= 2$

Solid ball

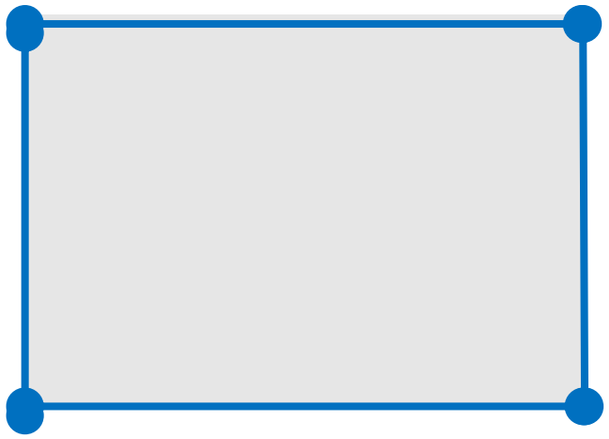
$EC = N - E + F - V$

$= 8 - 12 + 6 - 1$

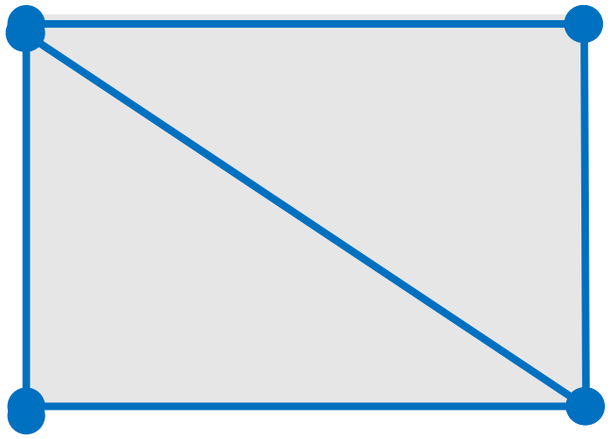
$= 1$

Computing Euler characteristic by parts

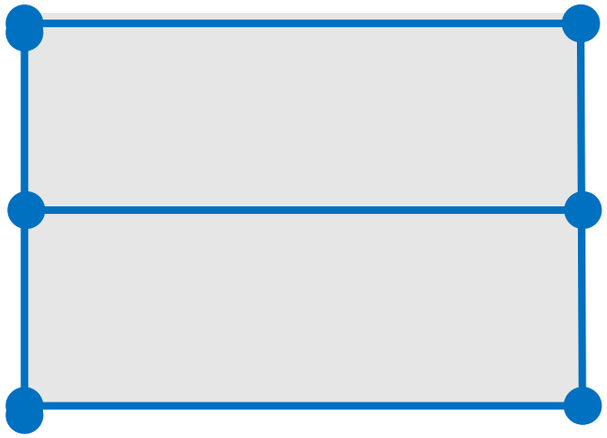
Cover an object with polyhedrons



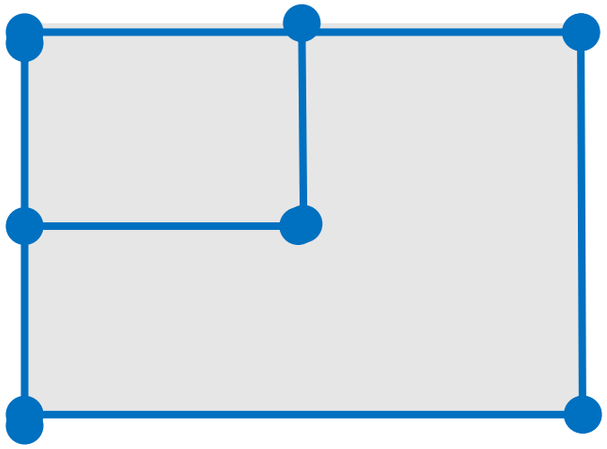
EC = N – E + F
=4 – 4 + 1
= 1



EC = N – E + F
=4 – 5 + 2
= 1



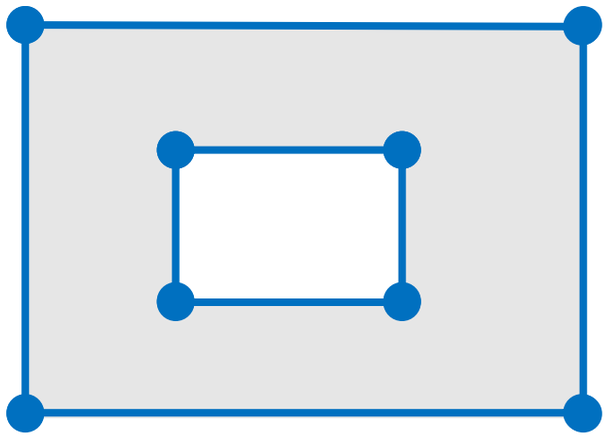
EC = N – E + F
=6 – 7 + 2
= 1



EC = N – E + F
=7 – 8 + 2
= 1

Computing Euler characteristic by parts

Cover an object with polyhedrons

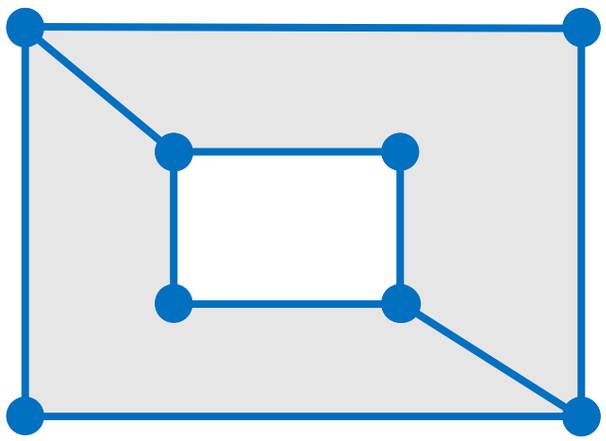


Incorrect computation

EC = N – E + F

=8 – 8 + 1

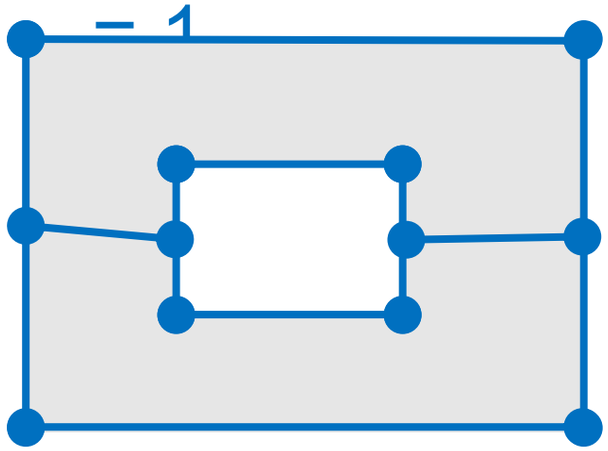
– 1



EC = N – E + F

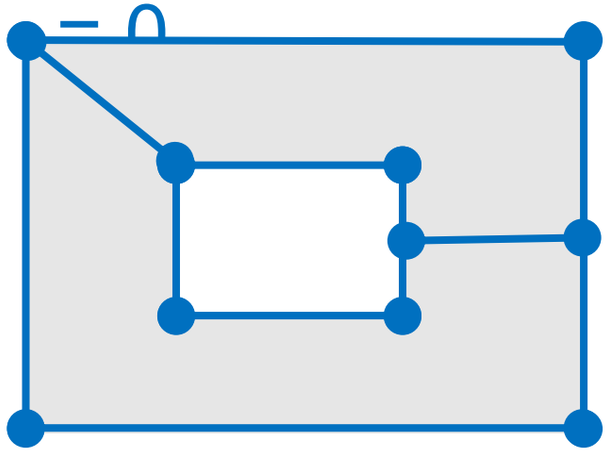
= 8 – (8+2) + (1+1)

– 0



EC = N – E + F

=(8+4) – (8+6) + (1+1)

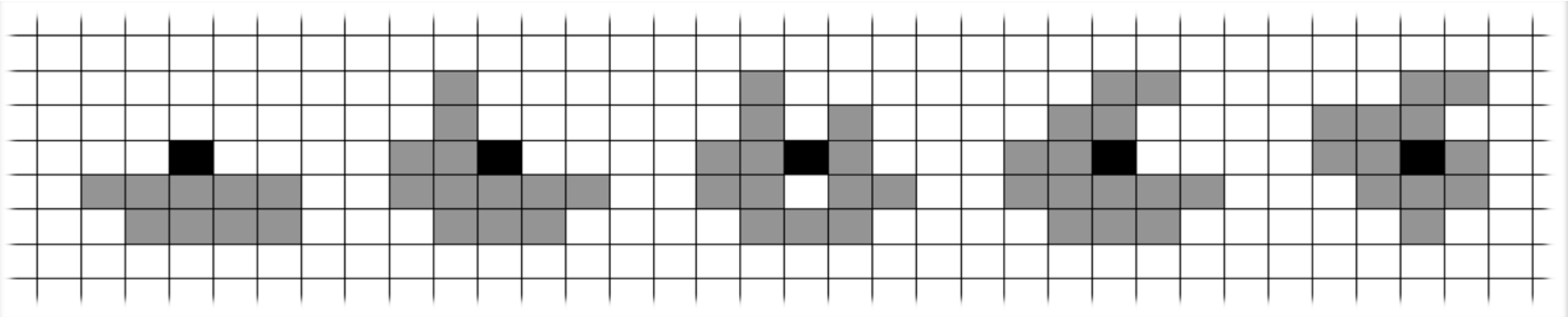


EC = N – E + F

=(8+2)–(8+4) +(1+1)

= 0

Iterative computation (online) of Euler characteristi



# neighbors	1	2	2	3	4
# Vertex	2	1	0	0	0
# Edge	3	2	2	1	0
# Face	1	1	1	1	1
EC change	0	0	-1	0	1

Question: Given binary image, write an iterative algorithm

Computing Euler characteristic in 3D image

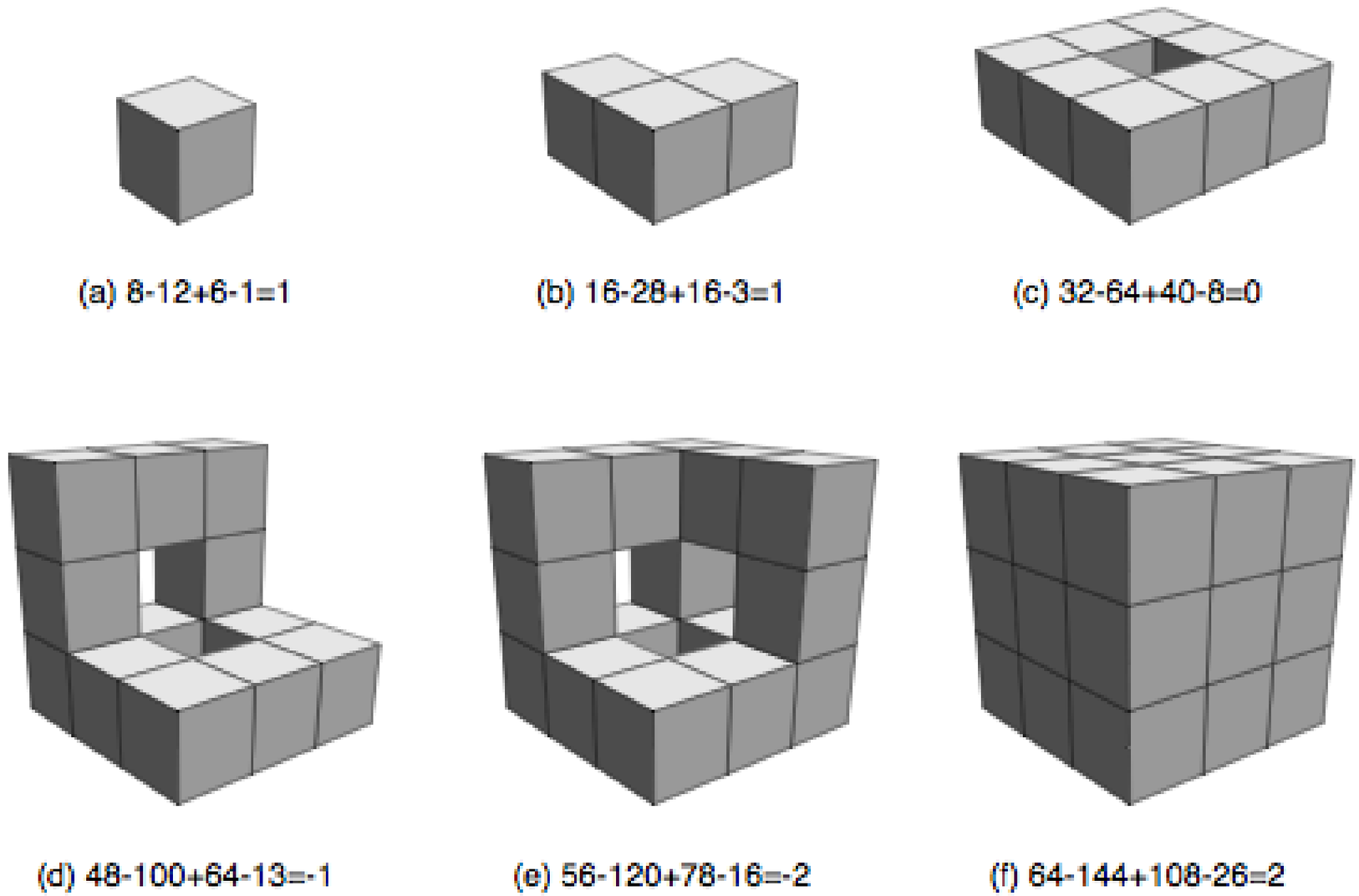
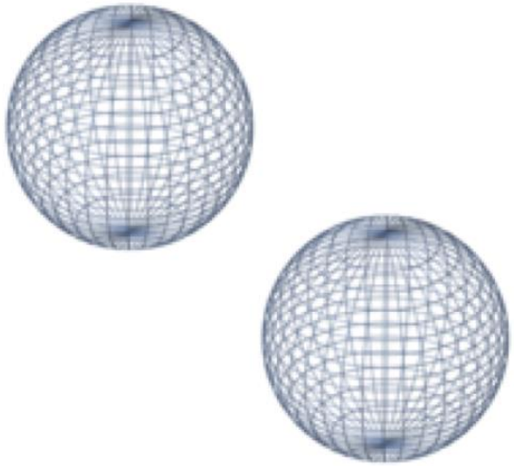


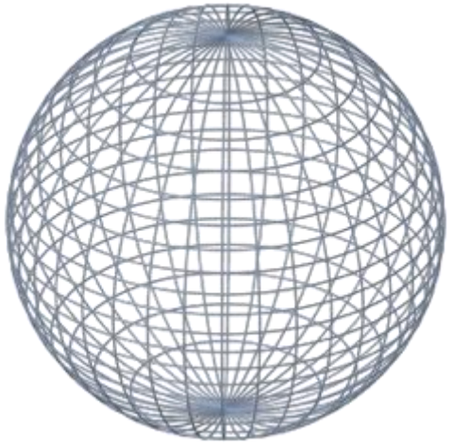
Image credit: Keith Worsley

Partition search region into voxels.
EC = # vertices - # edges + # faces - # volume

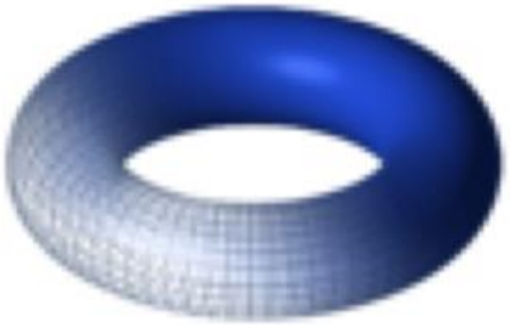
Euler characteristic: most widely used topological invariant



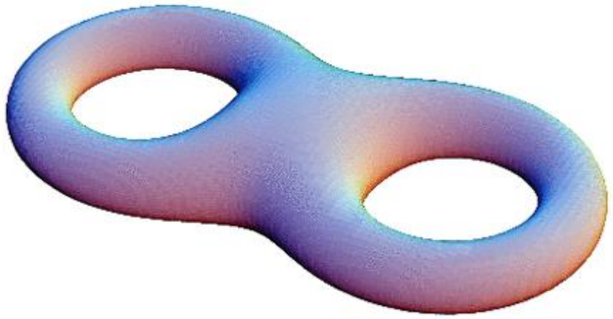
4



2



0



-2

For an object with n-handles, $EC = 2-2n$

*Question: Can you p*rove the statement?

Expected Euler characteristic/Betti numbers

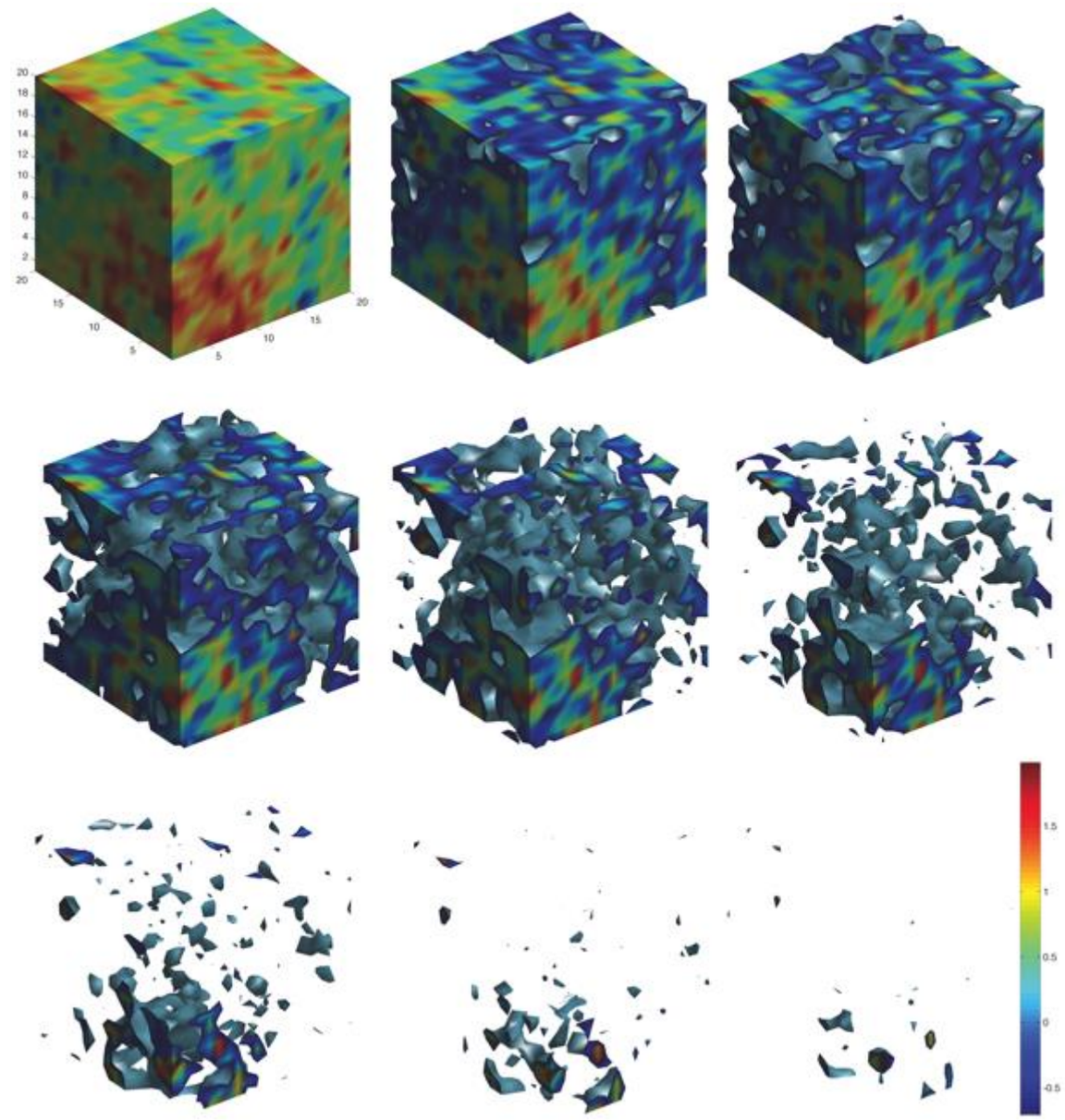
Random field, stochastic process

$$P\left(\sup_{x\in\mathbb{M}} T(x) > h\right)$$

$$A_h = \{x \in \mathbb{M} : T(x) > h\}$$

$$P\left(\sup_{x\in\mathbb{M}} T(x) > h\right) = \mathbb{E}\chi(A_h)$$

$$\chi(A_h) = \sum_j (-1)^j \beta_j(A_h)$$



Milor 1963 Morse theory
Adler, 1994 The geometry of random fields
Worsley et al., 1996 Human Brain Mapping

Betti numbers β_i

of i-dimensional holes/loops

β_0 = # of connected components = 3

β_1 = # of cycles = 1

Euler characteristic: $\chi = 3 - 1 = 2$

numbers β_i # of i-dimensional holes/loops

β_0= # of connected components = 3

β_1= # of 1D holes = 1

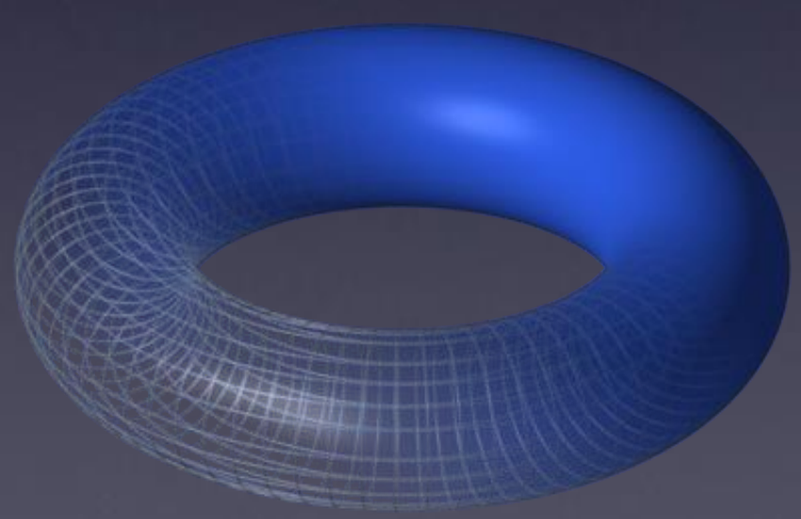
β_2= # of 2D cavities =0

Betti-number representation:

(3,1,0,0,...)

Euler characteristic:

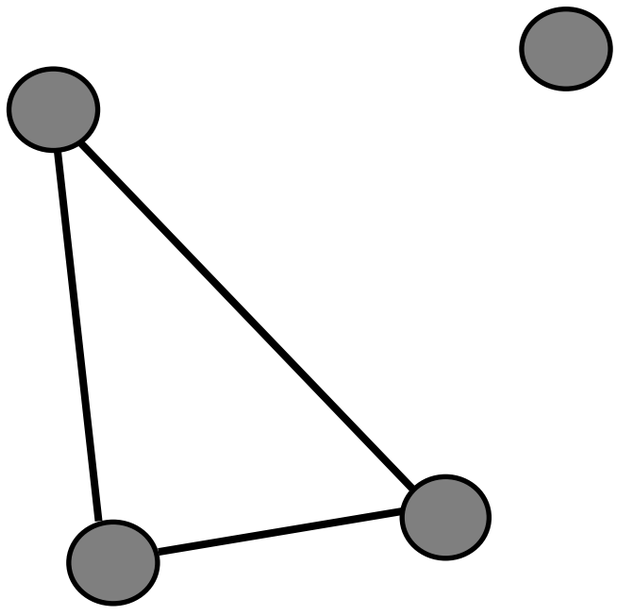
$$\chi = \beta_0 - \beta_1 = 2$$



$$\beta_0 = 1, \beta_1 = 2, \beta_2 = 1$$

(1,2,1,0,0,...)

Betti numbers in graphs and networks



$\beta_0 = 2$

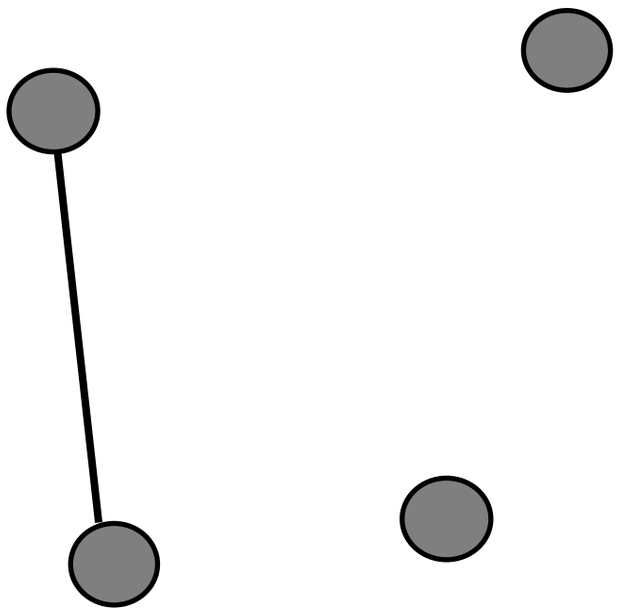
$\beta_1 = 1$

$$\beta_0 - \beta_1 = 1$$

$p = 4$

q = 3

$$p - q = 1$$



$\beta_0 = 3$

$\beta_1 = 0$

$$\beta_0 - \beta_1 = 3$$

$p = 4$

q = 1

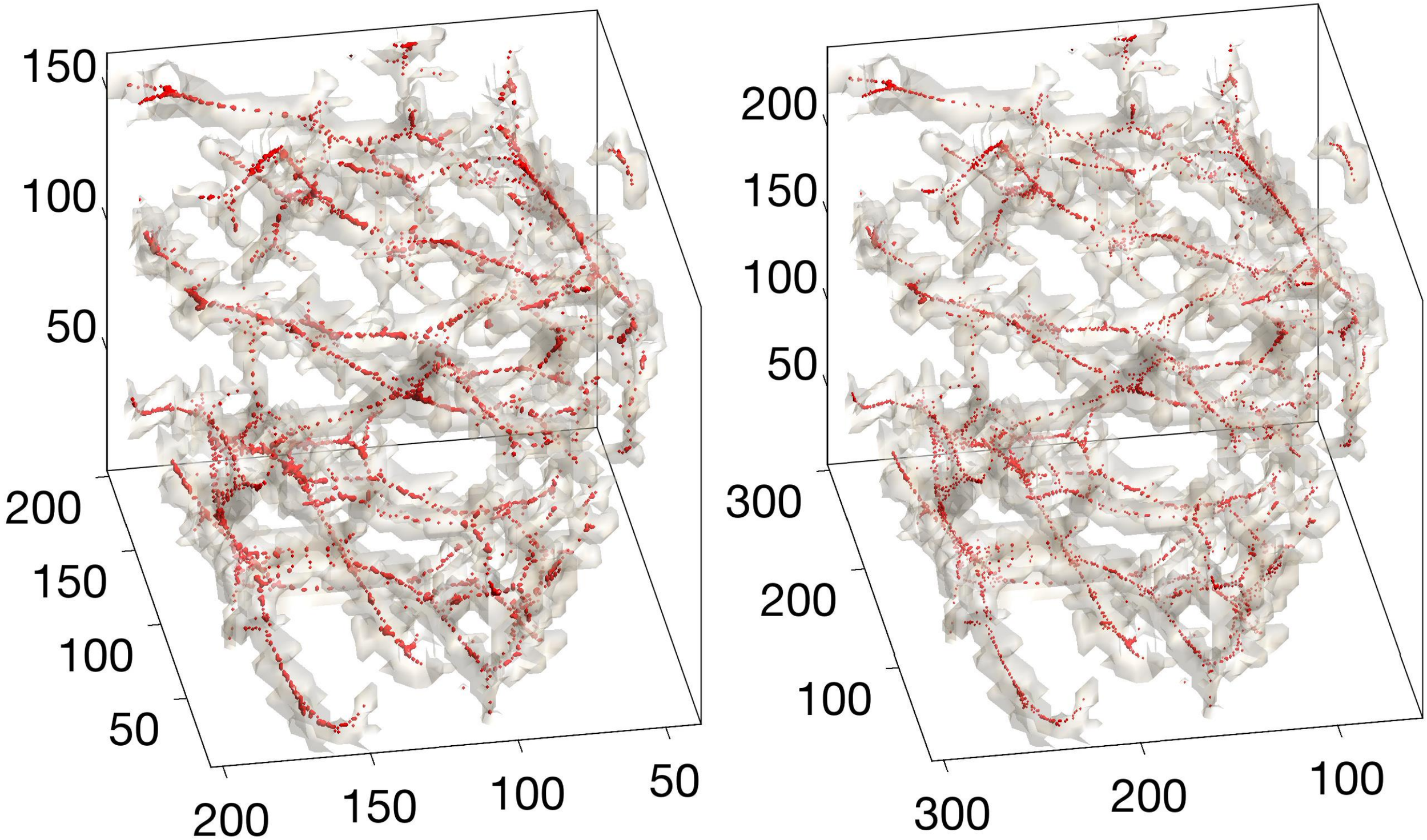
$$p - q = 3$$

How to compute the number of cycles in big network data?

How many cycles in the network?

Tree data (1-simplices)

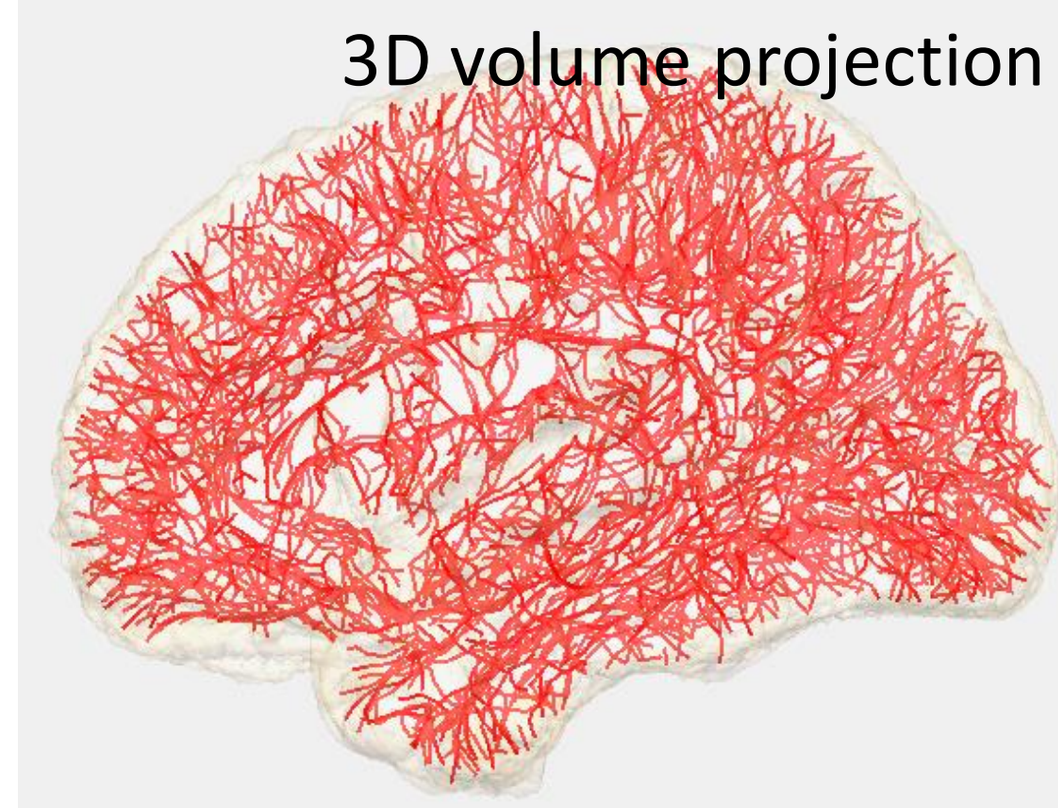
Lung blood vessel trees from CT



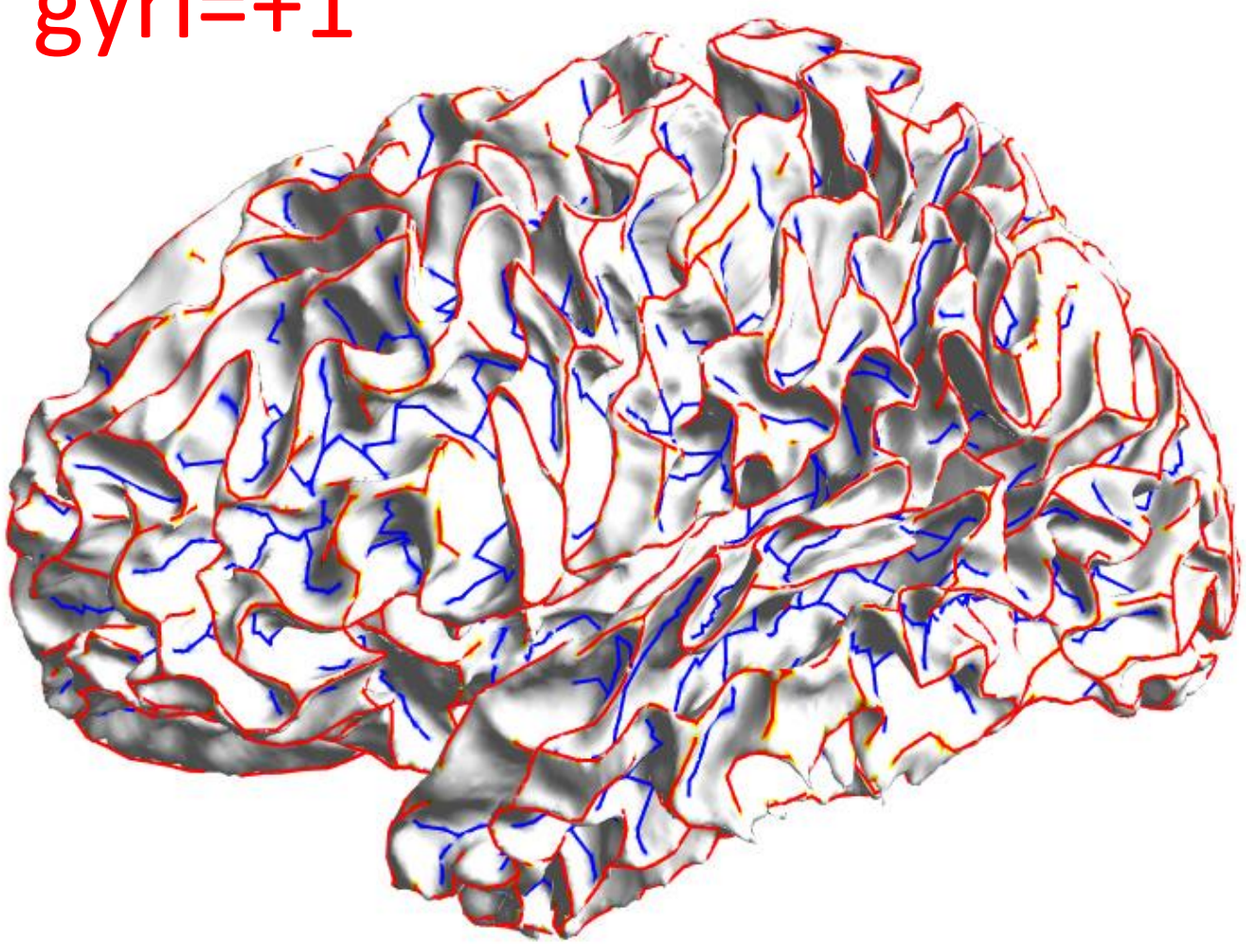
Chung et al. 2018 EMBC

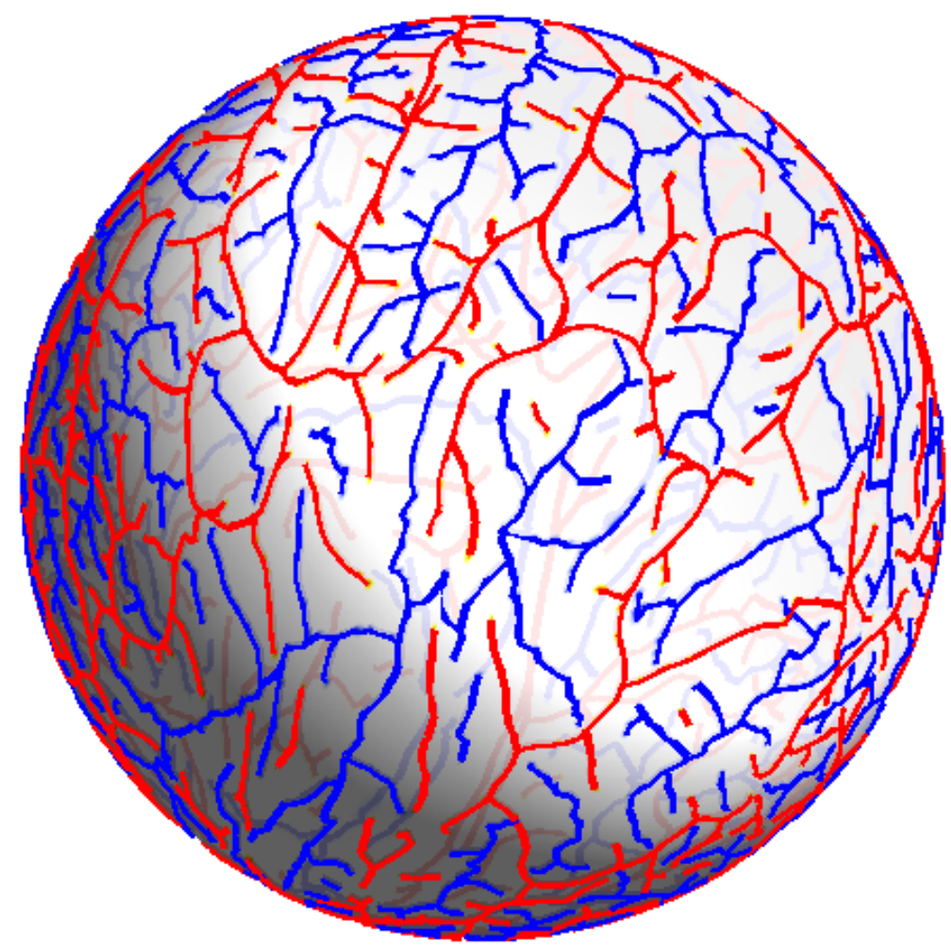
Chung et al. 2019, Mathematics of Shapes and

Sulcal and gyral trees of brain from MRI



sulci = -1
gyri=+1





White matter surface

Trees on manifold

Topology of tree

of edges = # of nodes -1

Euler characteristic
= # of edges - # of nodes
= -1

Betti numbers
Betti0 = 1, Betti1 = 0

$$\chi = \sum_{i=0}^{\infty} (-1)^i \beta_i = \beta_0 - \beta_1 + \beta_2 - \beta_3 + \cdots .$$

Surface mesh data (2-simplices)

Marching cubes algorithm

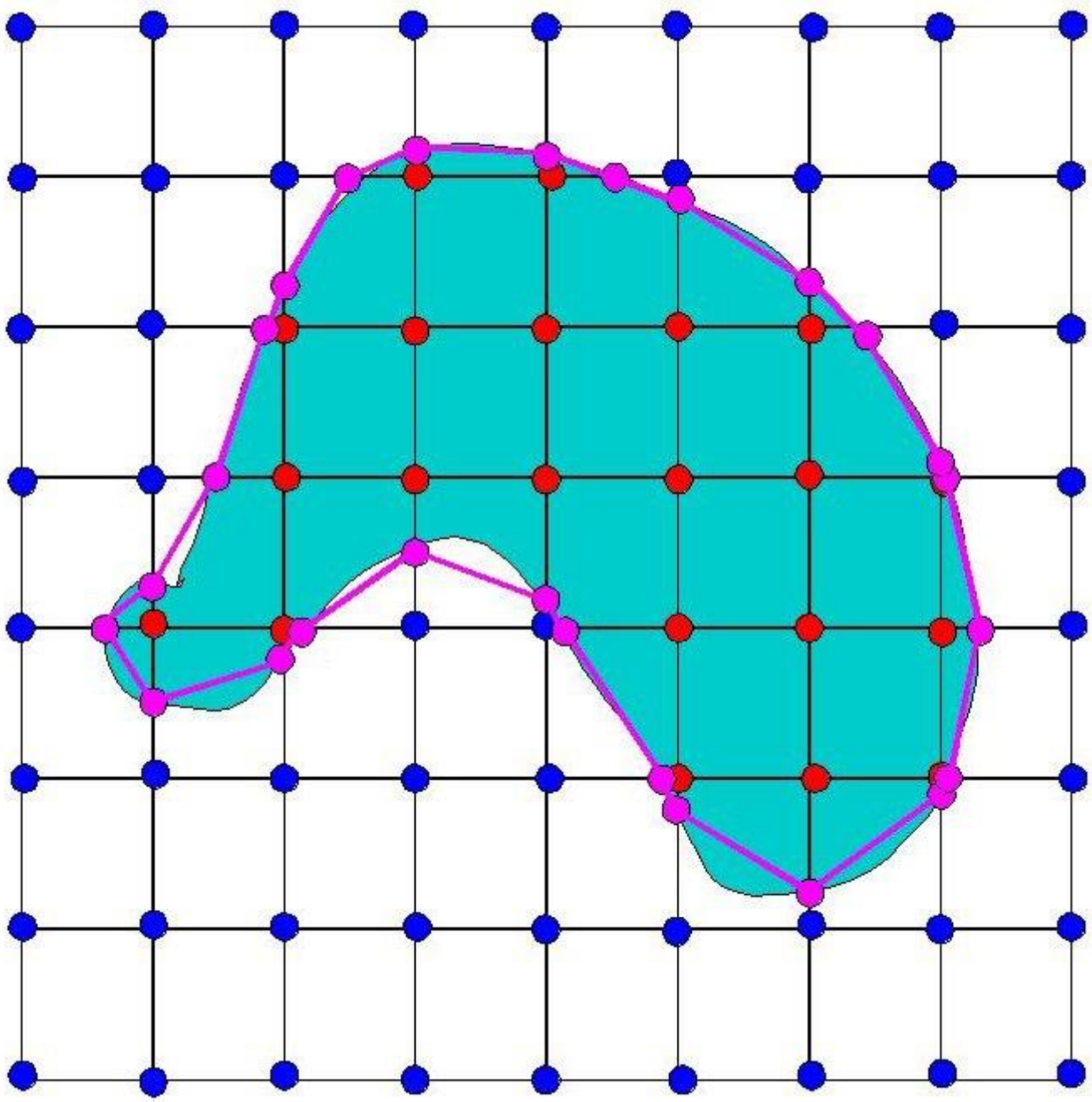
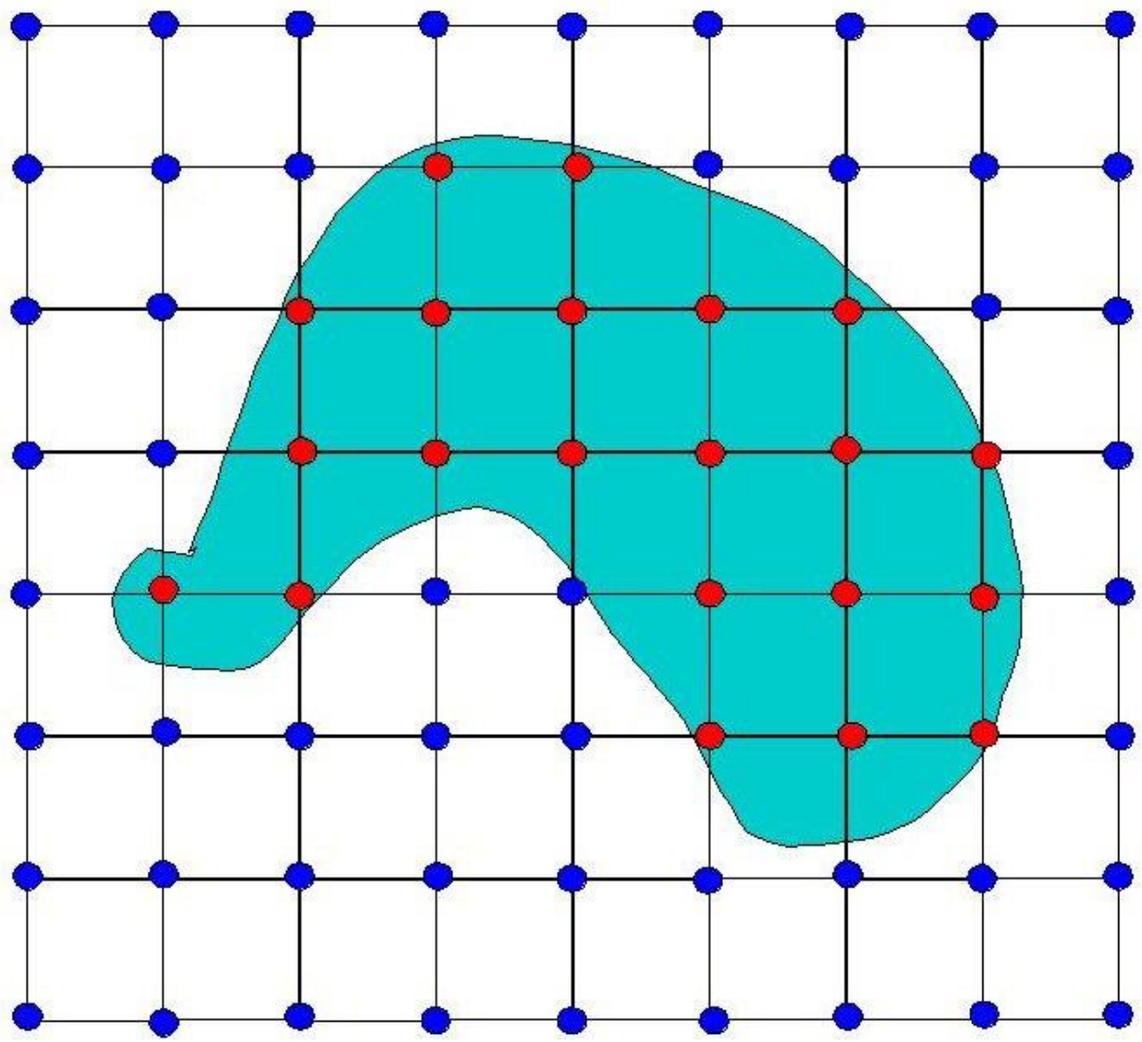
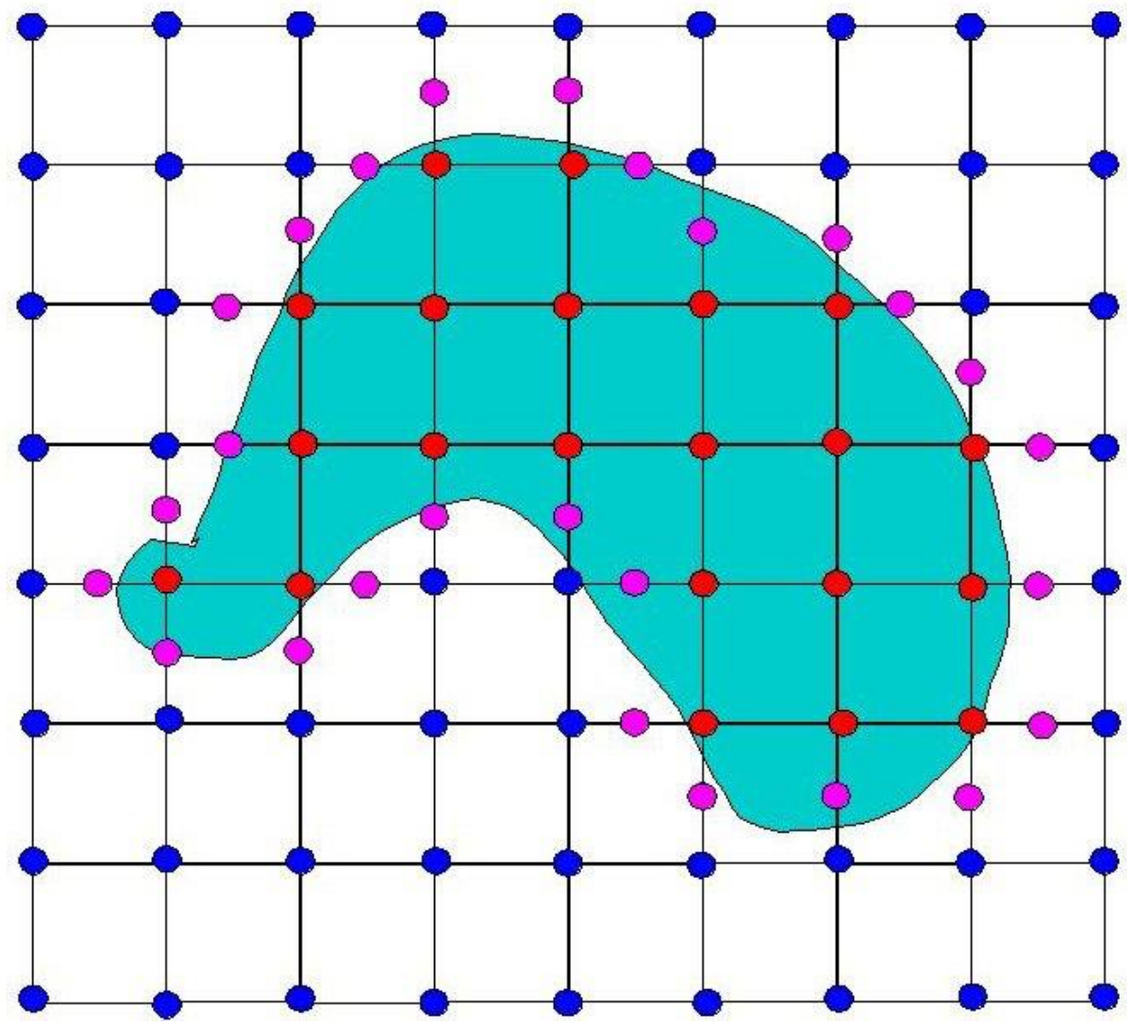


Image intensity values are classified into two parts: in & out

The marching cubes algorithm will extract the boundary of binary segmentation

Marching cubes algorithm





Depending on the amount of partial vacuuming, we connect the centers of cubes

Marching cubes algorithm in 3D

Cell consists of 8 voxel values:

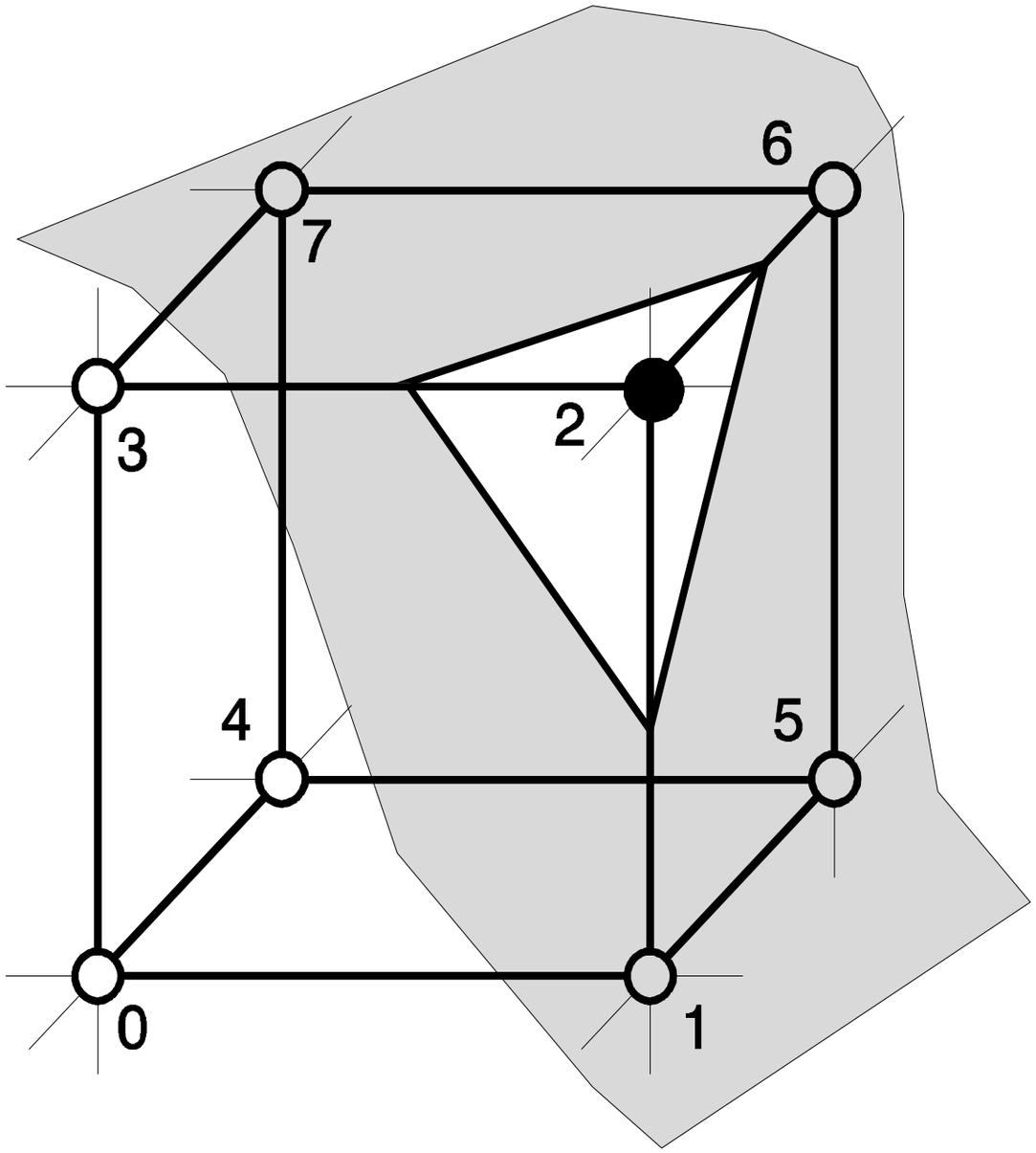
1. Classify each vertex as inside or outside

2. Build an index

3. Compute edge list from table[index]

4. Determine vertex coordinates

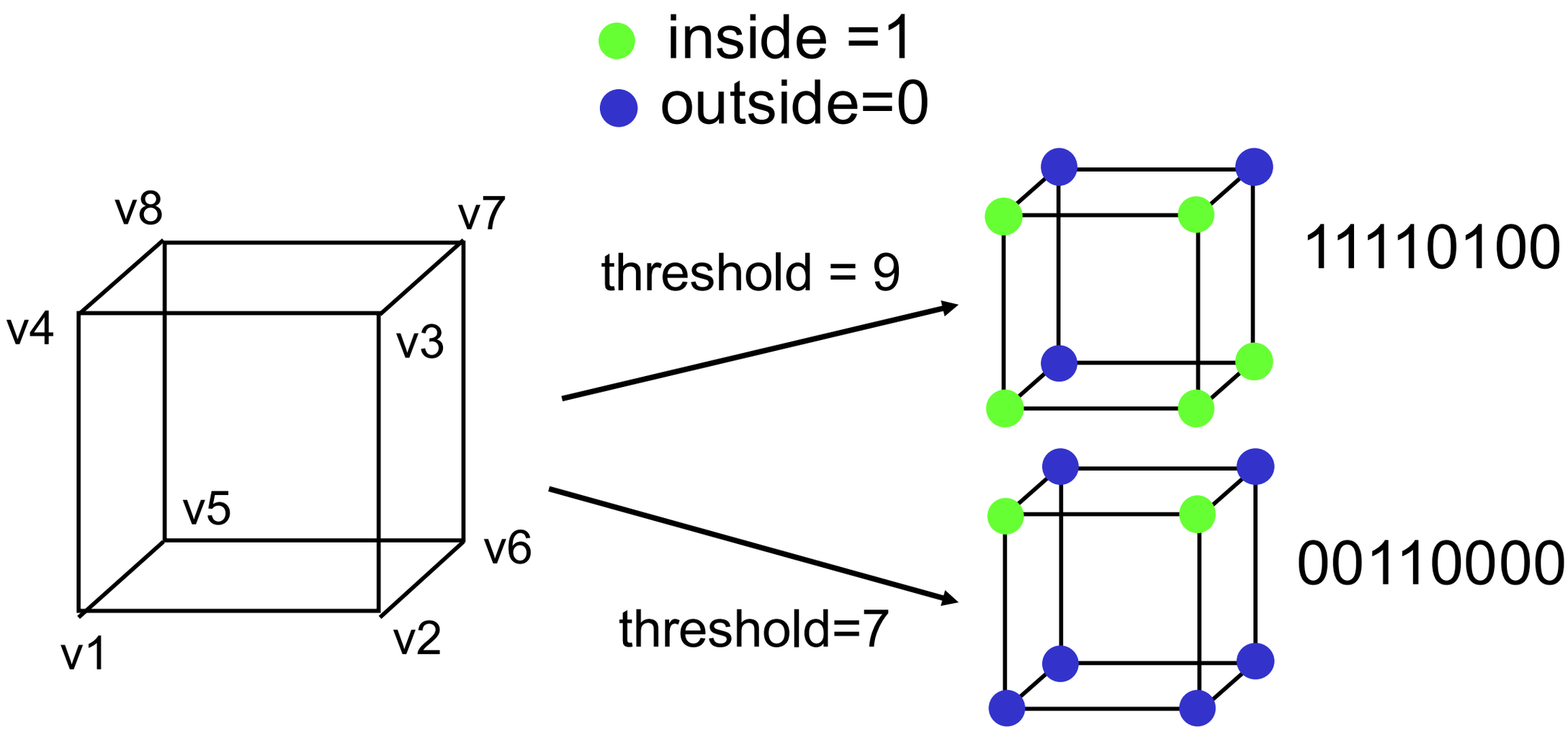
4. Go to next cell



0	0	0	0	0	1	0	0

Marching Cubes

Use the binary labeling of each voxel vertex to create an index



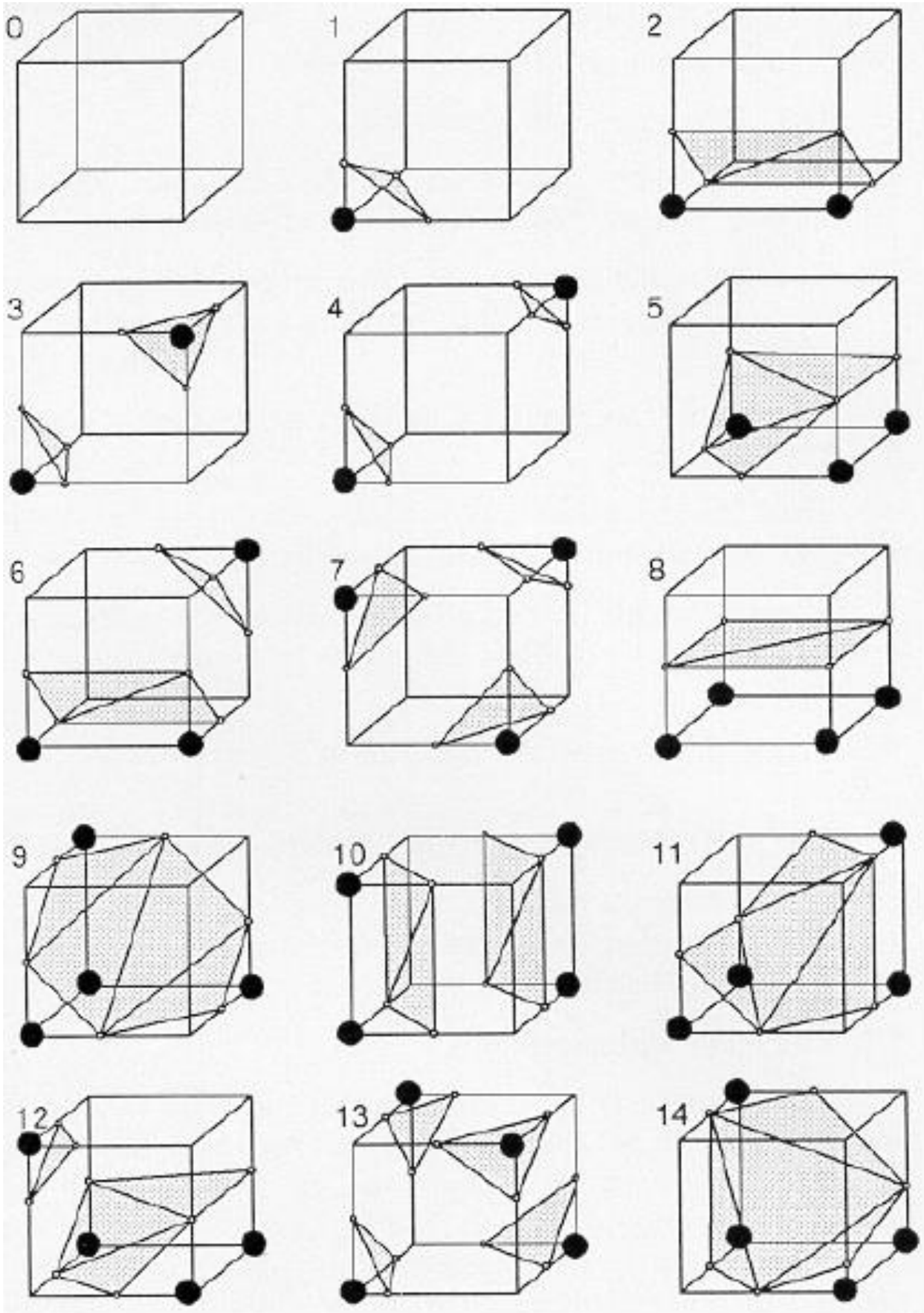
Index:

v1	v2	v3	v4	v5	v6	v7	v8

Marching Cubes

All 256 cases can be derived from 1+14=15 base cases due to symmetries

Invented by Carl Crawford (GE's Medical Systems Business Group). Expert on CT reconstruction.



Table

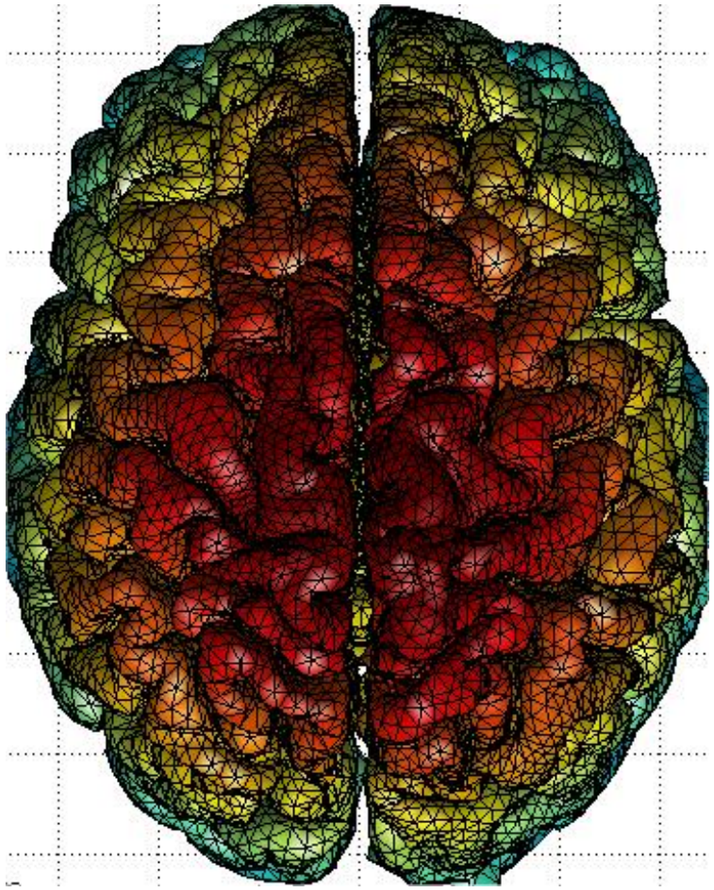
Marching cubes in MATLAB

```
FV =
isosurface(X,Y,Z,V,ISOVALUE)
```

computes isosurface geometry for data V at isosurface value ISOVALUE.

Optional: arrays (X,Y,Z) specify the coordinates at which the data V is given. The struct FV contains the faces and vertices of the isosurface and can be passed directly to the PATCH command.

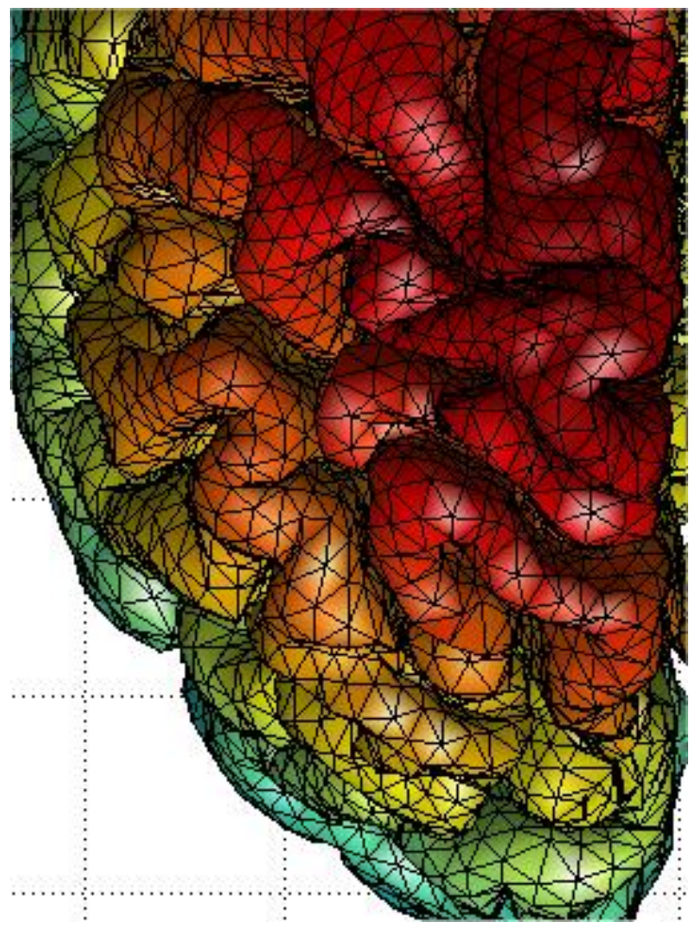
It is based on simpler version of marching cubes.

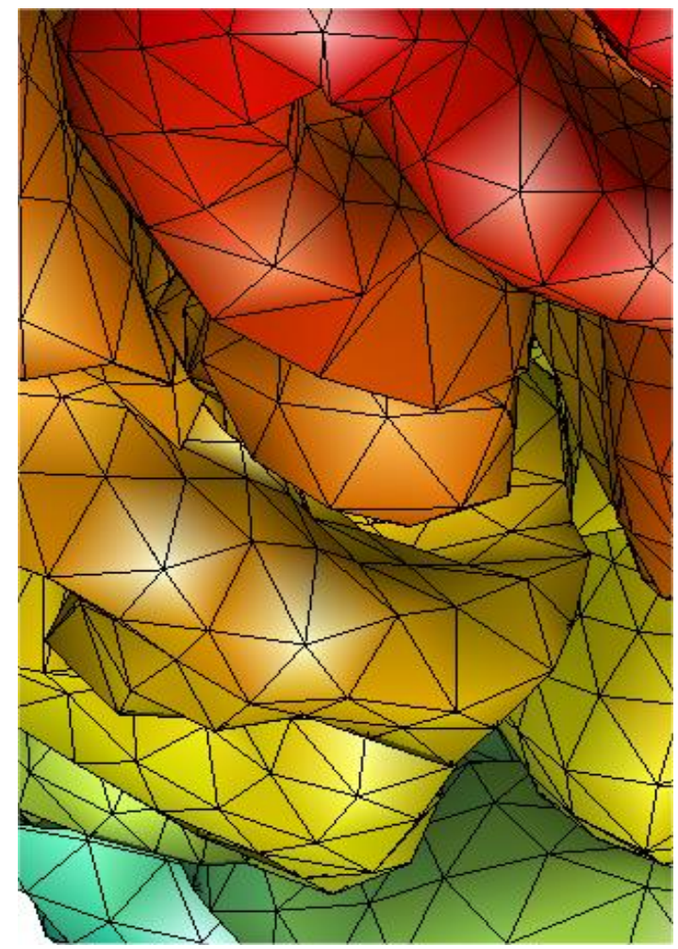


Output of isosurface.m

Triangle mesh data structure

Basis of most surface rendering tools for 3D computer games: as 3D Max Studio, Maya





Data structure for triangle mesh

```
>>surf =
    vertices: [1282x3 double]
       faces: [2560x3 double]
```

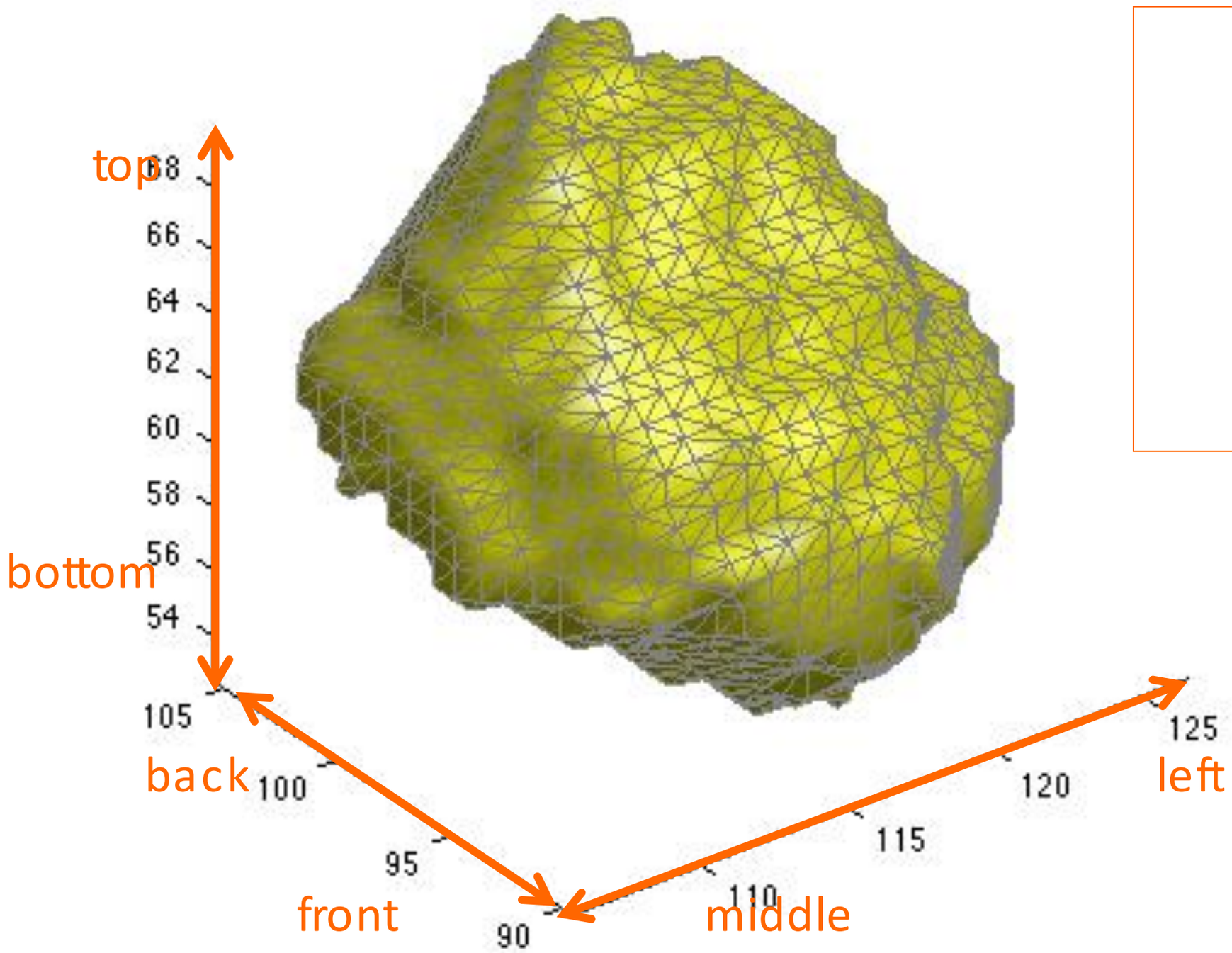
structured array

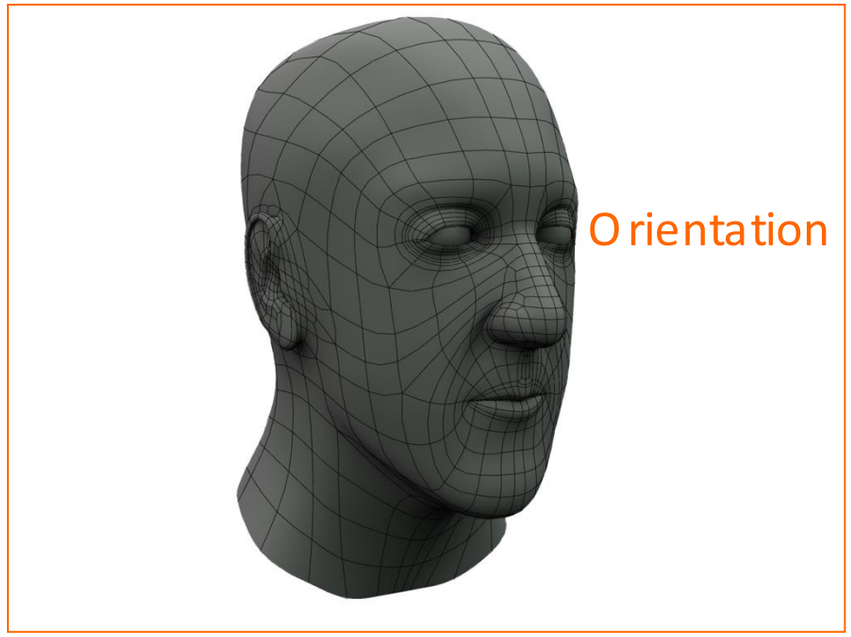
```
>>surf.faces
ans =
            1              2              3
            1              4              2
            1              3              5
           ...
```

```
>> surf.vertices
ans =
   75.0000    93.0000    51.5050
   74.5050    93.0000    52.0000
   75.0000    92.5050    52.0000
    ...
```

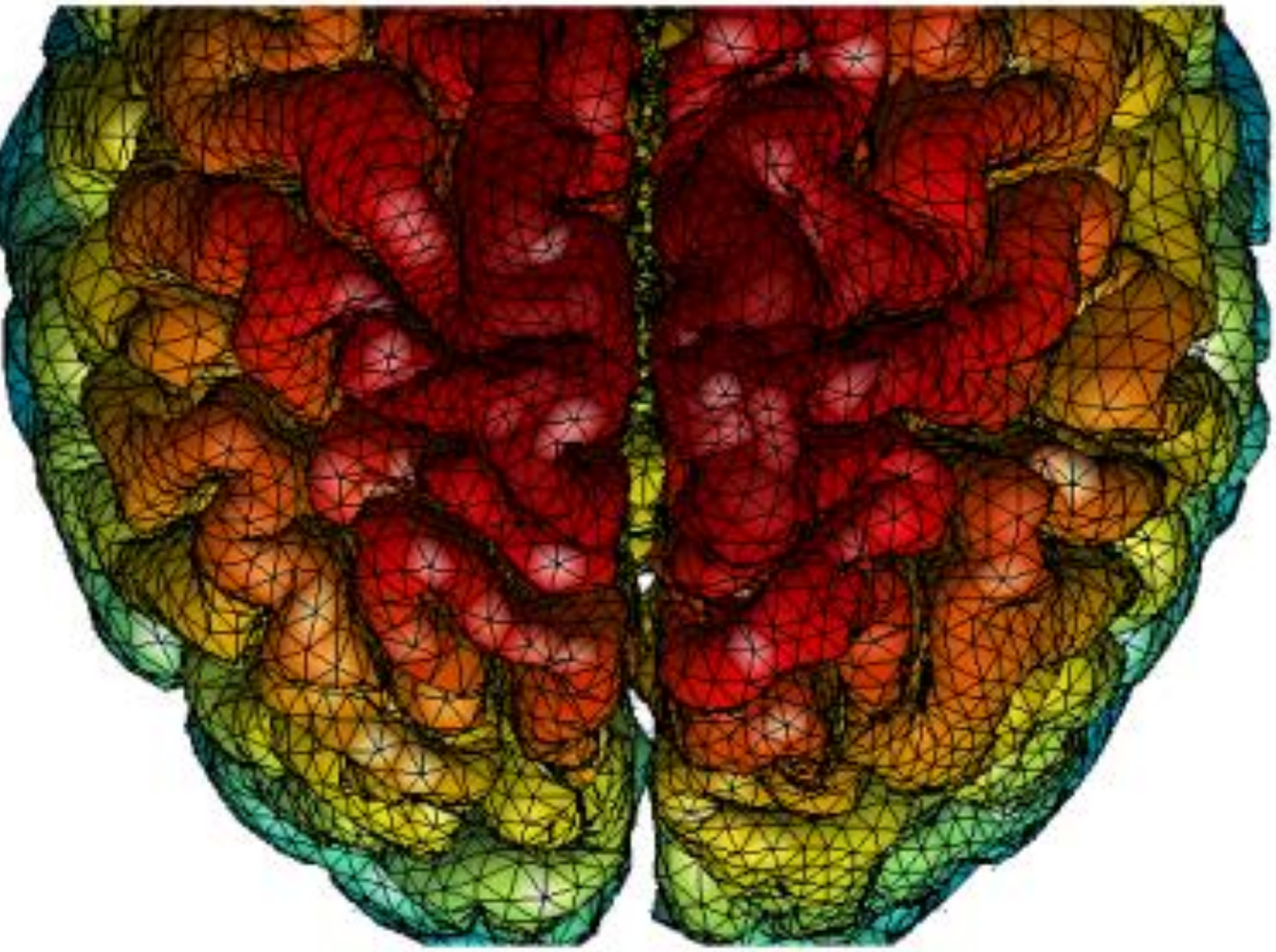
vertex coordinates

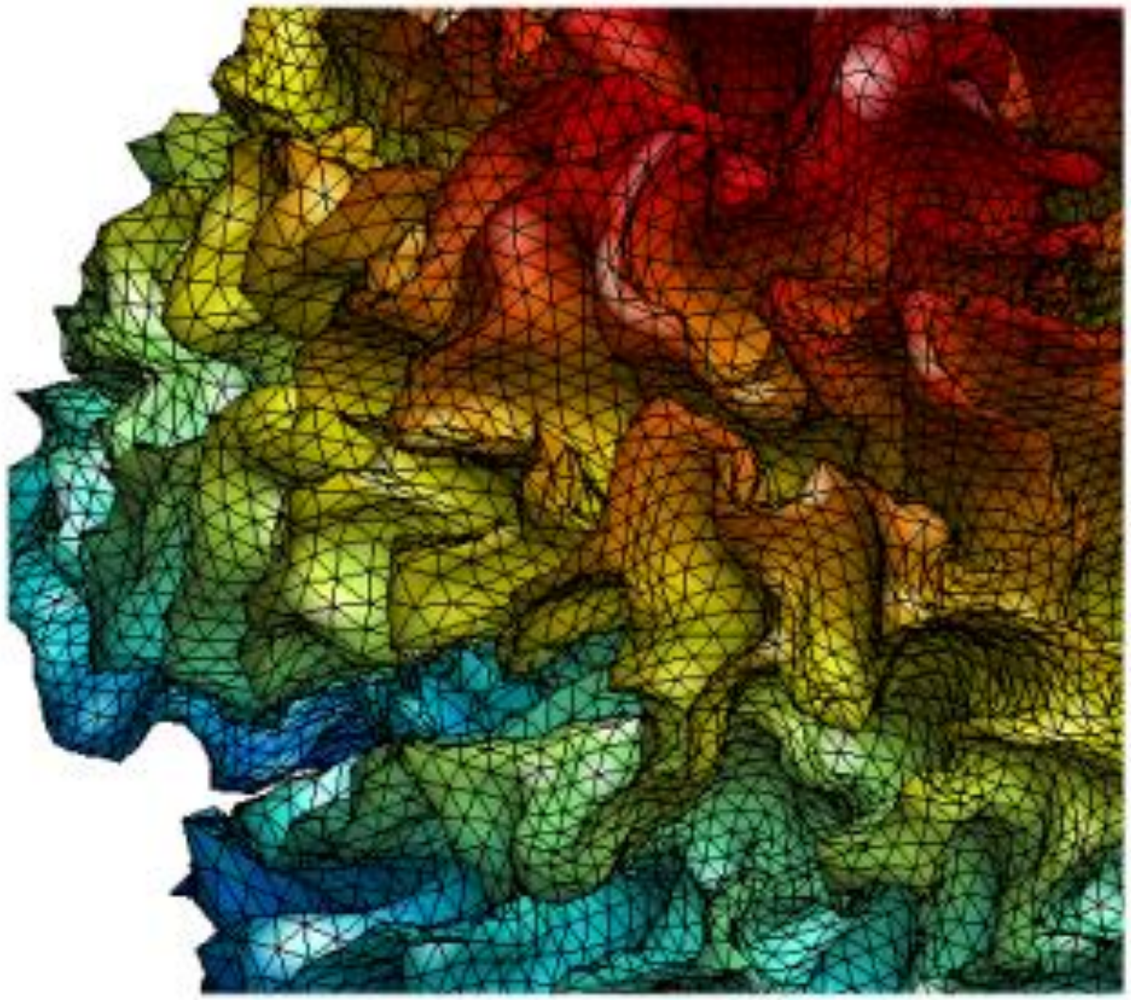
2D surface model of left amygdala using marching cubes algorithm





Matlab demo





Euler characteristic for a surface mesh

How to check the topologically defect of a surface?

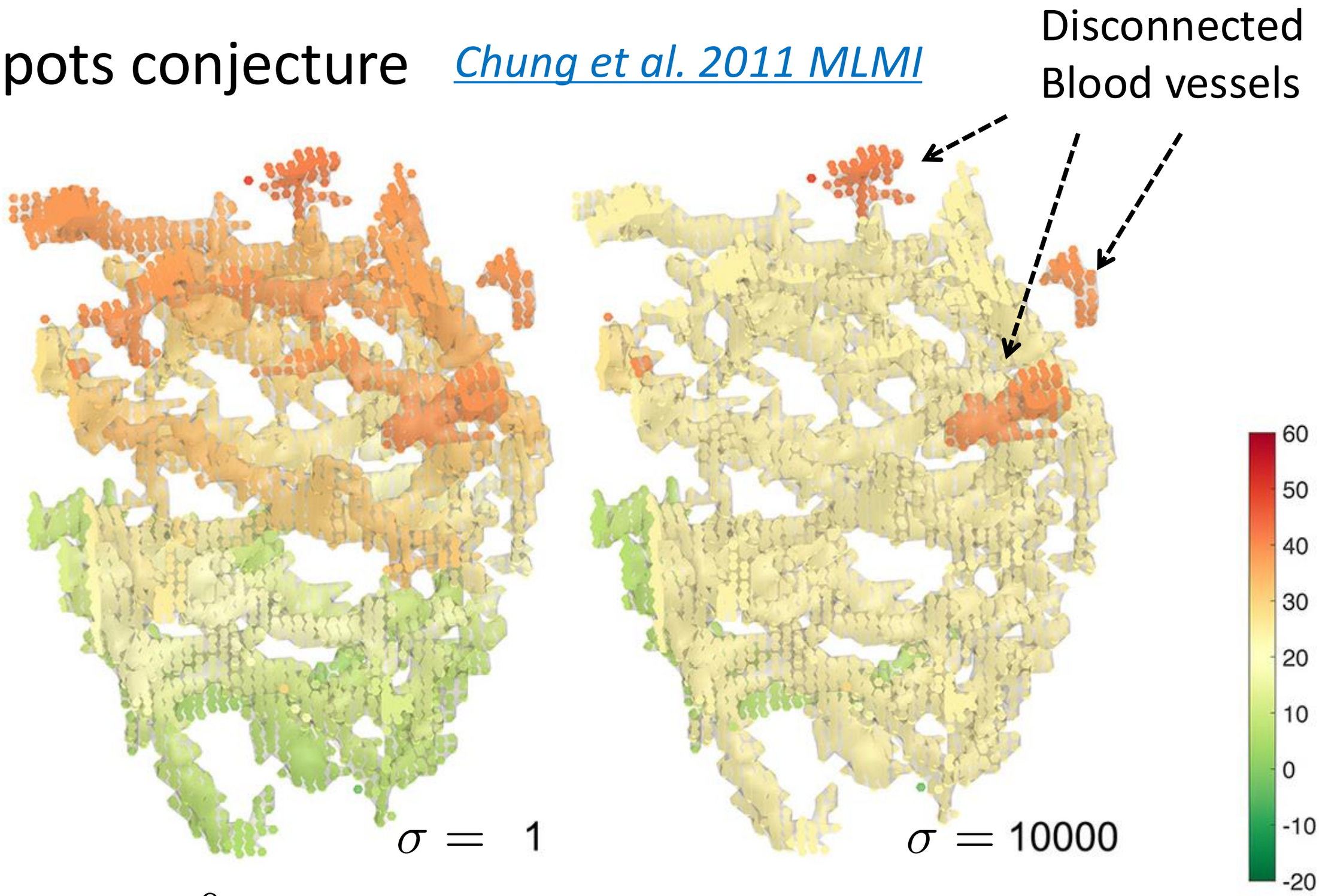
N - E +F = 2 for a surface topologically equivalent to a sphere.

For each triangle, there are three edges. Since two adjacent triangles share the same edge, the total number of edges is E = 3F/2. Hence, we have F=2N – 4 for a closed surface.

Advanced topology problems

Hot spots conjecture

Chung et al. 2011 MLMI



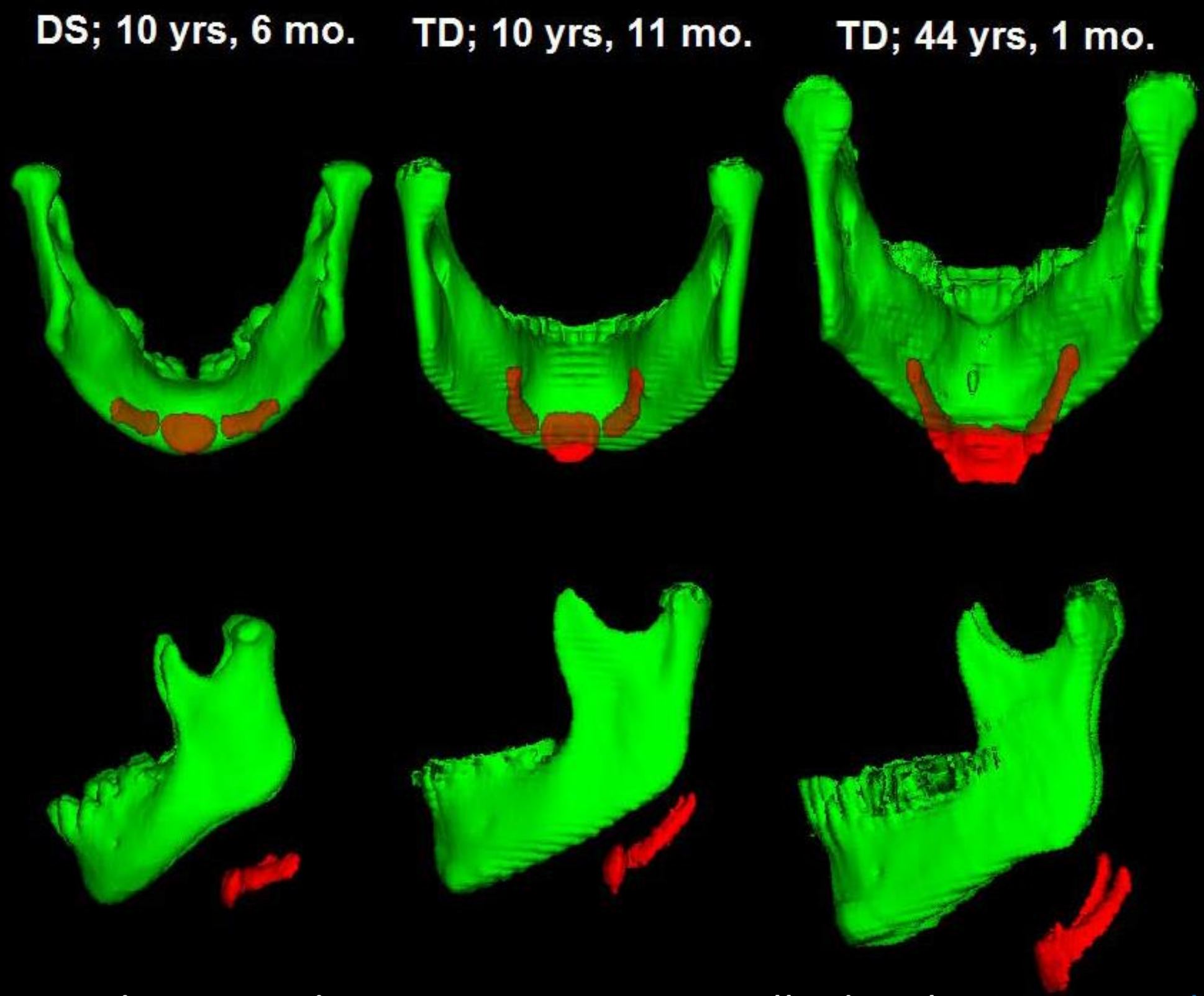
$$K_\sigma * f(p) = \frac{\int_{\mathcal{M}} f(p)\, d\mu(p)}{\mu(\mathcal{M})} + f_1 e^{-\lambda_1 \sigma} \psi_1(p) + R(\sigma, p)$$

Diffusion

Mean signal

Topology term

Topology changing bone fusion

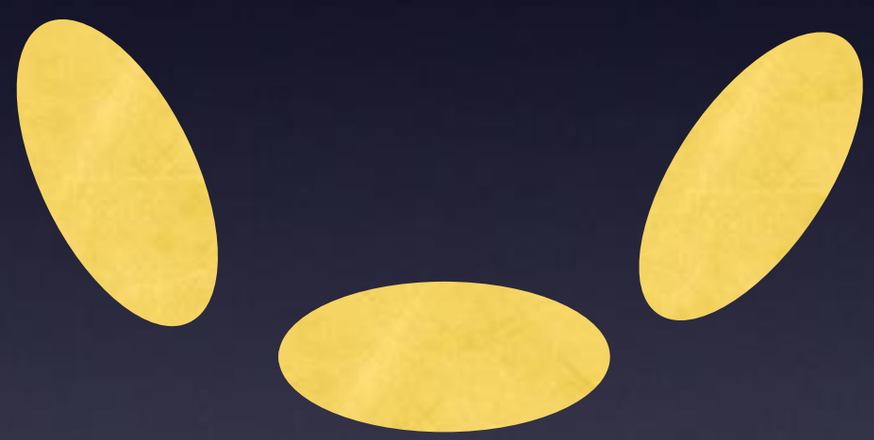


DS: down syndrome

TD: typically developing

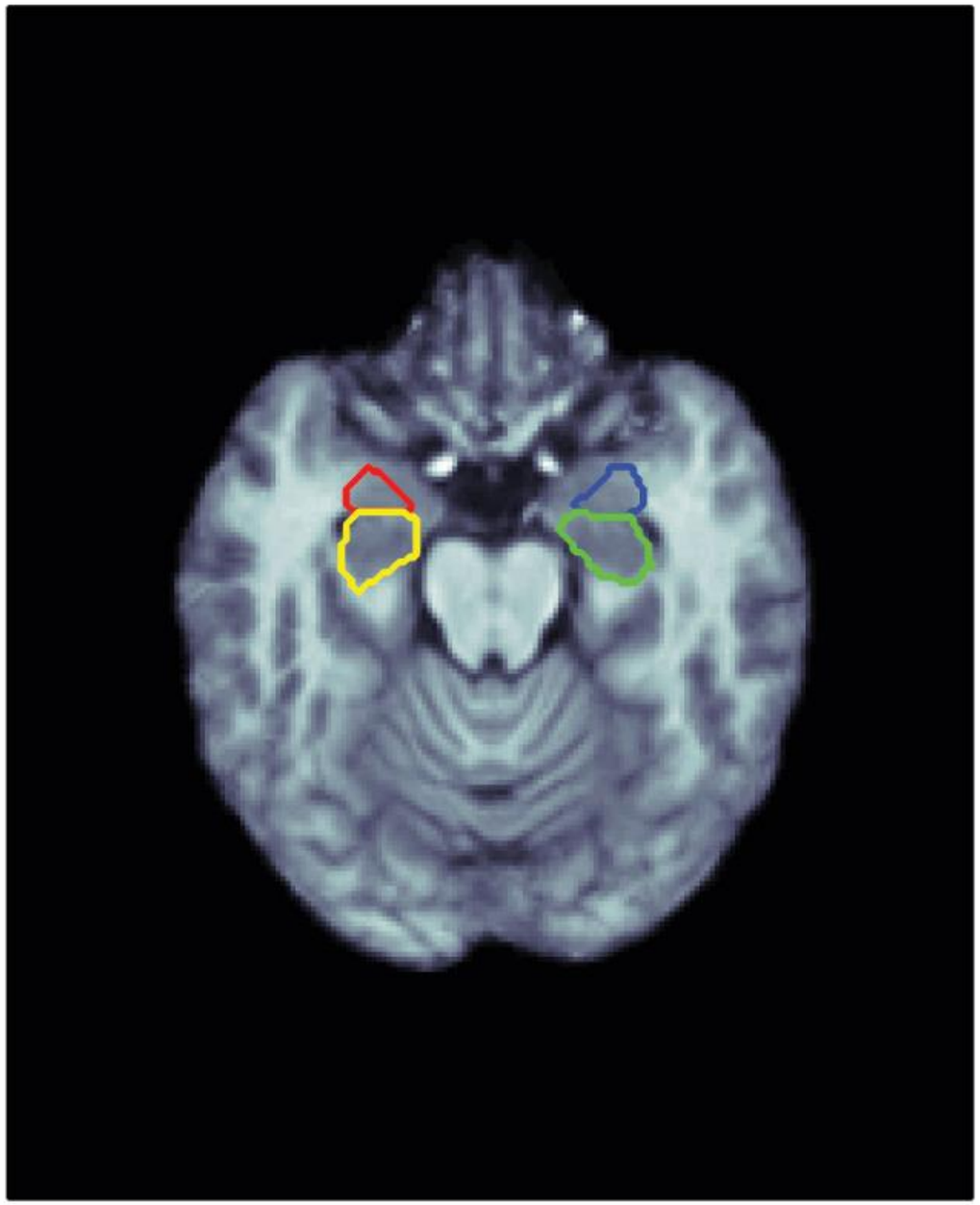
Chung et al. 2020

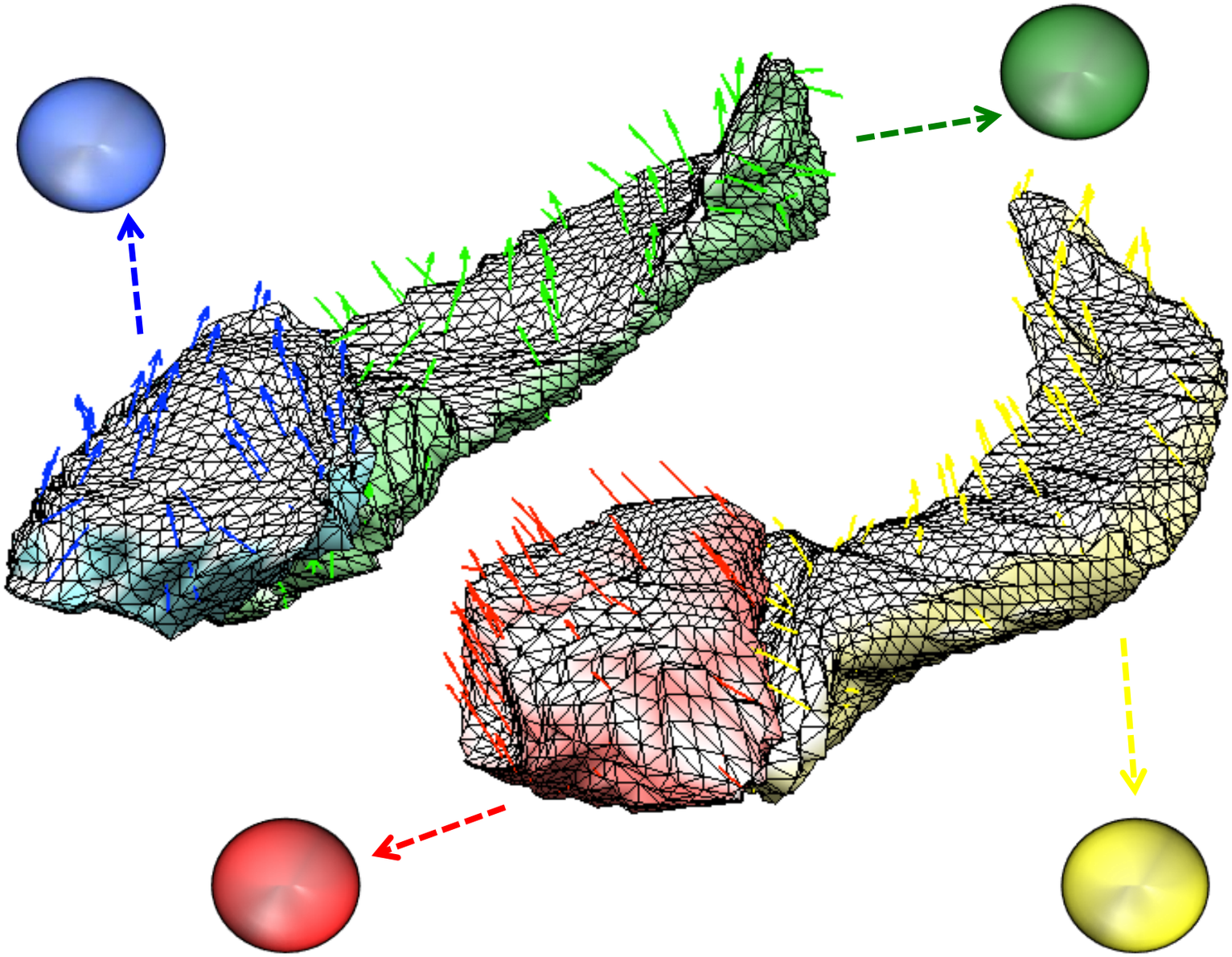
Is it possible to set up a coherent longitudinal growth model for topologically changing structures?



Project: This is nontrivial.

Spherical harmonics (SPHARM) parameterize with respect to a sphere (single connected structure)





How to represent 4 disjoint structures using Hyper-SPHARM basis:
<u>Hosseinbor et al. 2015 Medical Image Analysis</u>